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The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

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Title page

Title: The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

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Keywords

Sudden Infant Death Syndrome; Hearing screen; Otoacoustic emission; Auditory brainstem response; autopsy; auditory system.

Abstract

Introduction A theory has emerged, suggesting that abnormalities in the auditory system may be associated with Sudden Infant Death Syndrome (SIDS). However, the current clinical evidence has never been systematically reviewed.

Aims To systematically review the literature on the association between auditory system pathology measured by newborn hearing screening or autopsy with brainstem histology and Sudden Infant Death Syndrome (SIDS).

Methods Pubmed, Embase and Web of Science were searched systematically. Only human studies including a reference group were assessed. Studies were eligible for inclusion if they examined infants exposed to otoacoustic emissions, auditory brainstem response, or had autopsies with brainstem histology of the auditory system. SIDS was the primary outcome and the secondary outcome was either a near-miss SIDS episode (NMSIDS) or an apparent life-threatening event (ALTE). Due to high heterogeneity, a narrative synthesis was conducted. Risk of bias and quality of evidence was assessed using the Newcastle-Ottawa Scale and Grading of Recommendations Assessment, Development and Evaluation (GRADE).

Results Twelve case-control studies were included. Seven studies on otoacoustic emission (OAE) or auditory brainstem response (ABR) had a high degree of inconsistency. Contrarily, four out of five studies reporting on brainstem histology found that auditory brainstem abnormalities were more prevalent in SIDS cases than in controls. However, the quality of evidence across all studies was very low.

Conclusion This systematic review found no clear association between auditory system pathology and SIDS. The higher prevalence of histological abnormalities in the auditory system of SIDS may indicate an association. However, further studies of higher quality and larger study populations are needed to determine whether these findings are valid.

Article Summery

Strength and limitations of this study

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1. This is the first systematic review based on human studies that investigates the association between the auditory system pathology and sudden infant death syndrome (SIDS).
2. The systematic review was completed according to the guideline of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).
3. Risk of bias was assessed systematically with the validated Newcastle-Ottawa Scale for case-control studies.
4. Due to the heterogeneity of the included studies, this is a narrative systematic review without meta-analysis.

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Declaration of interest:

All authors have no conflict of interest.

Author Contributions

All authors had substantial contributions to the conception and design of the study, acquisition of data, or analysis of data. All authors were involved in drafting the article or revising it critically for important intellectual content. All authors approved the final version to be submitted.

Abbreviations

SIDS = Sudden infant death syndrome

NMSIDS = Near miss sudden infant death syndrome

ALTE = Apparent life-threatening event

OAE = Otoacoustic emission

ABR = Auditory Brainstem Response

SOC = Superior olivary complex

MSO = Medial superior olivary complex

IC = Inferior Colliculus

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Introduction

Sudden Infant Death Syndrome (SIDS) is defined as the sudden death of an infant under the age of one year occurring unexpectedly, with no other potential explanation of death and an autopsy compatible with SIDS.(1) The recommendation for infants to sleep in a supine position has significantly decreased the incidence of SIDS.(2) In the UK, the SIDS incidence decreased from 2.3 per 1,000 live births in 1988 to 0.77 per 1,000 live births in 1992 after the "back to sleep" campaign with similar trends being observed in other western countries.(3, 4) However, SIDS is still one of the leading causes of death in infancy (0.30 per 1,000 live births) and a scientific mystery with no explanation for the bereaved parents.(4) A near-miss SIDS (NMSIDS) episode may be considered as a precursor before an actual SIDS event. However, the link to SIDS is discussed vividly.(5-7) NMSIDS is also classified as an apparent life-threatening event (ALTE), which to some degree have replaced the term NMSIDS.(7) Previous studies have examined infants surviving NMSIDS episodes anticipating to improve the understanding of SIDS.(8, 9)

Even though SIDS is a diagnosis of exclusion, deficiencies in the cardiorespiratory system and arousal response are believed to play an important role in SIDS.(10-12) Human and animal studies have indicated that the inner ear is connected to the respiratory system and may be essential for the arousal response.(13-18) Furthermore, the anatomical proximity between the respiratory system and the auditory pathway in the brainstem imply that there may be synaptic connections.(19, 20) If abnormalities in the auditory system are associated with SIDS and may be captured by newborn hearing screening, we would be able to follow infants at risk and potentially reduce the number of SIDS cases further. Two methods are used for newborn hearing screenings: Otoacoustic Emission (OAE) (21, 22) and Auditory Brainstem Response (ABR) (23, 24). OAE examines hearing from the external ear to the cochlea, while ABR tests the entire auditory pathway reaching the auditory cortex. Furthermore, if abnormalities of the auditory pathway can be shown histologically, we would get one step closer to a biological understanding of the cause of SIDS.

Several studies have investigated this association. However, the current human evidence has never been compiled. This systematic review aims to provide a comprehensive overview on the literature on the association between auditory system pathology measured by newborn hearing screening or autopsy with histology of the brainstem and Sudden Infant Death Syndrome (SIDS). Additionally, NMSIDS and ALTE are included as secondary outcomes to investigate the possible association.

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Methods

This systematic review was conducted according to the guideline of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).(25) A protocol with predefined methods, search strategy, and inclusion criteria was registered at the International Prospective Register of Systematic Reviews (PROSPERO) 7 September 2020 (reg. number: CRD42020208045).

Study selection and eligibility criteria

Studies, languages, and population

Case-control and cohort studies of infants below one year of age were eligible for inclusion. Experimental studies and studies without a reference group were excluded. We included studies in English, Danish, and other languages with adequate translation by Google Translate.

Exposure

Auditory system pathology was defined by abnormal OAE, ABR, or autopsy of the auditory system with histology of the brainstem including the cochlear nuclei, the superior olivary complex (SOC), and the inferior colliculus (IC).(26)

Comparators

In case-control studies, SIDS cases were compared to survivors (ABR or OAE studies) or infants dying from other well-defined causes (autopsy studies). In cohort studies, infants with normal OAE or ABR served as the reference.

Types of outcome

The primary outcome was SIDS defined as the sudden death of a child under the age of one year that occurred unexpectedly and with an autopsy showing no other potential explanation of death.(1) Studies with no or other definitions were still included but rated accordingly. We included NMSIDS and ALTE as secondary outcomes. NMSIDS was defined as severe life-threatening apnoea, while ALTE was defined as a combination of apnoea, marked changes in muscle tone, colour change, choking or gasping, and an apparent need for resuscitation by vigorous stimulation or ventilation.(9, 27)

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Search strategy

The literature search was conducted in Pubmed, Embase, and Web of Science. A search strategy was developed for each database using words related to the auditory system and SIDS. Medical Subject Headings (MeSH) in Pubmed and the Explode function in Embase was used. References of included studies were scrutinized as well as citations identified in Scopus. The search strategy was developed by the reviewers and peer-reviewed by a scientific librarian. The search was performed on 8 September 2020. The complete search strategy is available in Additional file 1.

Study selection

The search results were transferred to Endnote X9©(28) and duplicates were removed. The search results were then uploaded to Covidence (29) to manage the study selection process. Titles and abstracts were screened and any articles that seemingly met the inclusion criteria were extracted for full-text analysis. This process was conducted independently by two reviewers (KD and MA) with any disagreements resolved by discussion or by a third reviewer (TBH).

Data extraction

Data extraction was performed by two reviewers (KD and MA). The data were extracted using a predefined template, which was tested before use to identify any missing data plots. Authors of eligible studies were not contacted regarding missing information. The following data were extracted from the included studies: 1) title, authors, year of publication, country, study design, and aim of the study; 2) characteristics of the study population including age, sex, gestational age at birth, birth weight, sleeping position and condition, maternal smoking, and socioeconomic status; 3) methods of hearing assessment and autopsy; 4) definition of SIDS, NMSIDS and/or ALTE; 5) point estimates and statistics; and 6) information for assessment of risk of bias and quality of evidence by the Newcastle-Ottawa Scale (NOS) and the Grading of Recommendations Assessment, Development and Evaluation (GRADE).

Risk of bias

A modified version of the NOS for case-control studies and cohort studies was used to assess the risk of bias.(30) The assessment was performed independently by two reviewers (KD and MA) and any disagreements were resolved by discussion or by a third reviewer (TBH). Each study was evaluated within three categories; the selection process, the comparability between groups, and the exposure. A maximum of nine points could be awarded to each study by NOS. “Low” risk of bias required 3-4

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points in selection, 1-2 points in comparability, and 2-3 points in exposure; "fair" risk required 2 points in selection, 1–2 points in comparability, and 2–3 points in exposure; and 'high" risk was given 0-1 point in selection or 0 points for comparability or 0-1 point in exposure. Regarding comparability, one point was awarded if the study adjusted or matched for sleeping position or condition, while an additional point was awarded if they adjusted for either maternal smoking, sex, gestational age at birth, or birth weight.

Synthesis

The heterogeneity between included studies was too high to allow for a meta-analysis. A narrative synthesis was therefore performed, comparing differences and similarities of the studies according to Ryan *et al.*(31) Any missing estimates from the studies were calculated by use of GraphPad Prism version 8.00 for MacOS.(32) Statistically significance was defined as a two-sided *p*-value less than 0.05 and a 95% confidence interval (CI) was used.

Quality of evidence

The quality of evidence by GRADE was assessed independently by two reviewers (KD and MA).(33) Five factors could lower the quality: risk of bias, inconsistency, indirectness, imprecision, and publication bias. Serious limitations in any of these five factors resulted in a downgrading of the overall rating. The limitations could either be serious or very serious and downgraded one or two points. Regarding risk of bias, the studies were downgraded if they achieved one or zero points in comparability. Furthermore, they were downgraded one point if they missed a point in the selection process and exposure, or downgraded two points if they miss two points or more in these categories. The overall quality was rated as high, moderate, low, or very low.(33)

Risk of bias across studies

Due to high heterogeneity, a formal test of asymmetry could not be performed. Therefore, a qualitative analysis of publication bias was conducted with consideration of variability in study sample sizes. Furthermore, selective outcome reporting was assessed by comparing outcomes reported in the method and result section of the included studies.

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Results

Study selection

A total of 461 studies were identified by the search strategy. After the removal of duplicates, 302 studies were screened by title and abstract. Subsequently, 33 studies were screened by full text and a total of 12 case-control studies were included.(34-45) No cohort studies or studies with ALTE as outcome were identified. The selection process is documented in a PRISMA flowchart in figure 1.(25) Additional file 2 contains an overview of excluded studies with reasons for exclusion.

Study characteristics

Table 1 shows the main characteristics of each study including study period, country, population, outcome definition (SIDS and NMSIDS), methods for examination of the auditory system, and the provided estimations. The studies included a total of 335 cases and 392 controls. A total of 209 SIDS cases (34-40) and 126 infants with an NMSIDS episode (41-45) were included. OAE or ABR was assessed in 193 cases and 327 controls.(34, 35, 41-45) Histological examination of the brainstem was evaluated in 142 cases and 65 controls.(36-40) Three Italian studies of Lavezzi *et al.* (36-38) may have overlapping populations. These studies examined auditory system pathology of the brainstem in infants who died of SIDS.

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Table 1

Overview of study characteristics from 12 included case-control studies on auditory system pathology and sudden infant death syndrome (SIDS) and near-miss SIDS (NMSIDS)

Study	Inclusion period	Country of study population	Study population - How cases and controls are matched.	SIDS and/or NMSIDS definition	Method	No. cases No. controls	Results and effect estimates
				SIDS studies			
Blair, 2019 (34)	2007-2017	UK	Cases: Request was sent to families to contact The Lullaby Trust.(46) Controls: St. Michael's Hospital, Bristol and Birmingham Women's Hospital. Controls that survived the first year of life. Controls matched on sex.	*SIDS*: Each case was categorised according to the Avon clinicopathological system (47) as an unexplained death.	OAE	Cases: 36 Controls: 177	Non-statically significant difference between cases and controls. *p-values* > 0.05
Rubens, 2008 (35)	1993-2005	USA	Cases: infants born in the state of Rhode Island who died of SIDS. Controls: individually matched controls that survived the first year of life. Controls matched by sex, full term vs. preterm and NICU vs. routine nursery admission.	*SIDS* confirmed by autopsy	OAE	Cases: 31 Controls: 31	Statically significance difference between cases and controls at 2, 3 and 4 kHz on the right ear. *p-values* < 0.05
Lavezzi, 2015 (36)	N/R	Italy	Cases and controls: 69 infants were sent to Research Center and diagnosed according to the application of the guidelines stipulated to the Italian law. Controls: dead of well-defined causes. Controls matched on: sex.	*SIDS*: The sudden unexpected death of an infant <1 year of age, with onset of the fatal episode apparently occurring during sleep, that remains unexplained after a thorough investigation, including performance of a complete autopsy and review of the circumstances of death and the clinical history.(1)	Histology of cochlear nuclei, medial superior olivary nucleus (MSO) and inferior colliculus (IC)	Cases: 49 Controls: 20	Abnormal auditory brainstem nuclei: 47% SIDS cases and 10% ($p = 0.003$) - OR = 8.0 [CI: 1.7;38.1] Cochlear nuclei: 39% cases - 0 controls MSO: 24% cases - 10% controls IC: 41% cases - 0 controls

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Lavezzi, 2015 (37)	N/R	Italy	Cases and controls: 44 infants were sent to the Research Center and diagnosed according to the application of the guidelines stipulated by Italian law. Controls: dead of well-defined causes. Controls matched on: sex.	*SIDS*: no precise definition. Though, the Italian law decrees that, all infants suspected for SIDS who died suddenly in Italian regions within the first year of age must undergo an in-depth anatomo-pathological examination.	Histology of IC and superior olivary complex (SOC). Immuno-histochemistry for serotonergic fibers.	Cases: 29 Controls: 15	IC: 59% cases - 0 controls. $p < 0.05$ SOC: 38% cases - 7% controls. $p < 0.01$ - OR=8.6 [CI: 1.0;74.4] Serotonergic immunoreactivity: reduced or absent in 66% cases and none controls. $p < 0.01$
Lavezzi, 2012 (38)	N/R	Italy	No information Controls: dead of well-defined causes.	No definition	Histology of MSO	Cases: 30 Controls: 12	MSO alterations: 40% cases - 17% controls. - OR=3.3 [CI: 0.6;18.0]
Oehmichen, 1982 (39)	N/R	Germany	No information Controls: dead of well-defined causes. Controls matched on: sex and sleeping position.	*SIDS*: sudden and unexpected death with failure of autopsy, histologic evaluation and chemical-toxicologic analysis to demonstrate a definitive cause of death.	Histologic examination with immuno-histochemistry of IC	Cases: 11 Controls: 11	IC: no statistically significant difference concerning gliosis, neither astrocyte morphology or density. Microglia and lysozyme-positive cells were not observed.
Rickert, 2004 (40)	N/R	Germany	Cases: recruited through the German Study on Sudden Infant Death (GeSID). Controls: Controls: dead of well-defined causes. Controls matched on: sex	*SIDS*: The post-mortem examination, including histology, is by definition, negative with no anatomical or toxicological cause for the death ascertained.	Histologic examination with immuno-histochemistry of IC	Cases: 23 Controls: 7	IC: increased c-jun expression compared with controls. $p = 0.008$.
				NMSIDS studies			
Gupta, 1981 (41)	March 1980 to august 1980	USA	Cases: all infants suspected of NMSIDS were hospitalized on Stanford University Hospital. Controls: recruited in the Palo Alto community and from infants delivered at Stanford University Hospital. Controls matched on: sex	*NMSIDS* defined by strict criteria.(48) Episodes during sleep, in which parents believe that infants almost died. Cyanosis, apnoea and loss of consciousness was observed.	ABR	Cases: 9 Controls: 9	9/9 normal BAEP Interpeak latencies, wave latencies, interear latency were for all cases $p > 0.1$.
Kileny, 1982 (42)	N/R	Canada	Cases: infants found in their homes who required immediate resuscitation. Controls: normal control infants Controls matched on: sex and birth weight	*NMSIDS*: Infants who had been apnoeic, blue, or pale in their homes who required immediate resuscitation and for whom no cause could be found for this episode following a detailed hospital investigation.	ABR	Cases: 11 Controls: 11	11/11 normal ABR No statistically significant difference observed in interpeak latencies. $p > 0.05$

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Orlowski, 1982 (43)	N/R	USA	No information Controls with similar age distribution.	*NMSIDS* = IAS: infants older than 42 weeks postconception and younger than 12 months of chronological age who have experienced a frightening event at home or in the hospital characterized by prolonged apnoea (>20 sec.) requiring vigorous stimulation or resuscitation and not explained by clinical investigation. *SIDS*: The sudden death of any infant or young child which is unexpected by history and in which a thorough post-mortem examination fails to demonstrate an adequate cause for death.	ABR	Cases: 36 Controls: 25	36/36 abnormal ABR Statistically significant difference for peak latencies I, III and V, and amplitude III. $p < 0.05$ 42% cases had bilateral ABR abnormalities. 47% cases had left sided ABR abnormalities. 11% cases had right sided ABR abnormalities.
Pettigrew, 1985 (44)	N/R	Australia	Cases: infants admitted to hospital following a period of prolonged apnoea during apparent sleep. Controls: referred for study by verbal contact or had returned to hospital for regular postnatal check-up. Controls matched on: sex and prematurity.	*NMSIDS*: a period of prolonged apnoea during apparent sleep which was associated with pallor or cyanosis and which ceased only after vigorous shaking or mouth to mouth resuscitation.	ABR	Cases: 63 Controls: 67	88%: normal ABR. 12% abnormal ABR Difference in distribution of V-IIn interpeak interval. $p < 0.05$. Difference in distribution of V-I interpeak interval. $p > 0.05$.
Stockard, 1982 (45)	N/R	USA	No information Controls: normal infants matched on sex.	*NMSIDS*: episodes of prolonged apnoea during sleep in a previously thriving 3-weeks-to-6-month-old infant which result in generalized cyanosis and flaccidity and required resuscitation. *SIDS*: the unexpected, unexplained sudden death during sleep of a previously thriving infant between 3 weeks and 6 months of age which occurred in the absence of clear predisposing factors or discernible precipitants of cardiopulmonary arrest.	ABR	Cases: 7 Controls: 7	7/7 normal ABR 71% cases had I-V interpeak latency 1.2-2.2 SD above normal mean value.

Table 1: The study characteristics including study period, country, study population, outcome definition for SIDS and NMSIDS, methods for examination of the auditory system, and the provided results. Seven SIDS studies and five NMSIDS studies.
OAE: otoacoustic emission, ABR: auditory brainstem response. N/R: not reported. OR: odds ratio. CI: 95% confidence interval

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SIDS, NMSIDS, and controls

Seven studies (34-40) evaluated SIDS cases. All SIDS diagnosis was confirmed by autopsy in all infants. However, SIDS diagnosis were described differently across studies. Two studies (34, 35) included controls with infants who survived the first year of life, while five studies (36-40) included infants dying of well-defined causes. These well-defined causes were described further in four studies (36, 37, 39, 40) Five studies (41-45) investigated infants under the age of one year with an NMSIDS episode. Similarly, the NMSIDS episodes were described in varying detail; however, all studies defined it as an apnoeic episode that required resuscitation. The majority of NMSIDS studies found no clear predisposing or explainable cause for the NMSIDS episode.(41-43, 45) Stockard *et al.* (45) restricted the age criteria from three weeks to six months of life. Orlowski *et al.* (43) used the term Infant Apnea Syndrome (IAS) for NMSIDS. All NMSIDS cases were age-matched with healthy control infants.

The study results

OAE

OAE was applied as the exposure in two studies.(34, 35) The studies used various measures of OAE and different cut-offs for failed test. Rubens *et al.* (35) examined the signal to noise ratio (SNR) on both ears and discovered that SIDS cases had lower SNRs on the right ear compared with the left ear, which was reverse for controls. SNRs on the right ear were approximately 4 dB lower at 2, 3, and 4 kHz. An abnormal transient evoked otoacoustic emissions (TEOAE) on the right ear was more prevalent in SIDS cases compared with controls. Due to the result of Rubens *et al.* (35), Blair *et al.* (34) assessed the association between an abnormal OAE on the right ear and SIDS. They found no association between abnormal OAE and SIDS. SNRs were marginally higher rather than lower on the right ear among SIDS cases compared with controls, but not significantly.(34)

Histology

Histological and histochemical examination of the brainstem nuclei was performed in five studies.(36-40) Four studies (36-38, 40) found that abnormalities in the auditory structures of the brainstem occurred more frequently in infants with SIDS compared with infants with well-defined cause of death. All studies used hematoxylin-eosin staining. The three studies of Lavezzi *et al.* (36-38) performed a histological examination, which was carried out independently by two blinded pathologists. Lavezzi *et al.* 2015 (36) discovered hypoplasia of the cochlear nuclei, decreased number of neurons in the medial superior olivary nucleus (MSO), and cytoarchitectural abnormalities of the inferior colliculus (IC) in

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SIDS cases. Overall, 47% of SIDS cases and 10% of controls had two or more abnormalities in the auditory structures. According to this study (36), the odds of SIDS were 8.0 [CI: 1.7;38.1] times as high in infants with abnormal brainstem nuclei compared to infants with normal brainstem nuclei. Lavezzi *et al.* 2015 (37) and Lavezzi *et al.* 2012 (38) found similar abnormalities of the IC and MSO in SIDS cases compared with infants dying of well-defined causes. Furthermore, immunohistochemistry showed absent or very reduced markers of serotonin in the IC in SIDS infants.(37) Rickert *et al.* (40) investigated the expression of c-jun, an immediate-early-gene, which is part of the response to neuronal injury.(40, 49) The study discovered an increased expression of c-jun in the IC by immunohistochemistry in SIDS cases. Contrarily, the study of Oehmichen *et al.* (39) found no IC abnormalities in SIDS infants when semi-qualitatively evaluation of gliosis was performed.

ABR

Five studies (41-45) assessed the auditory pathway by ABR. Most infants were tested while asleep, but Orlowski *et al.*(43) and Pettigrew *et al.* (44) also allowed for chloral hydrate sedation and quiet rest. Various details of the ABR method were provided in the different studies. Similar ABR tests were mostly performed; however, tests differed by intensity (from 55 to 90 dB HL), use of masking (45), and electrode placement. Though, all infants within the same study were subjected to the same method. Two studies (41, 42) found no association between abnormal ABR and NMSIDS with all NMSIDS infants having normal ABR. Contrarily, Orlowski *et al.* (43) found that left-sided ABR abnormalities appeared more frequently among NMSIDS cases compared with controls. In the study of Pettigrew *et al.* (44), the majority (88%) of NMSIDS infants had normal ABR but approximately 12% had significant different interpeak intervals compared to controls. Stockard *et al.* (45) found normal ABR in all NMSIDS but found a tendency towards a prolonged V-I interpeak interval.

Risk of bias within studies

Table 2 shows the points awarded by NOS for case-control studies. The seven studies on SIDS (34-40) were classified as having *low risk of bias* with exception of Lavezzi *et al.* 2012 (38), who did not describe the selection process or the comparability between infants who died from SIDS and other well-defined causes. Furthermore, two studies (39, 40) did not state whether their histological examination was blinded. The five studies with NMSIDS cases (41-45) had *higher risk of bias*. Evaluation of exposure in all studies indicated little risk of bias. However, none of the studies adjusted for all our pre-defined confounders and two studies (38, 43) failed to control for any of them. Table 1

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shows which key variables were included in the matching for each study. In total six studies were assessed to having *low risk of bias* (34-37, 39, 40), while two studies (38, 41, 44) had a *fair risk of bias*, and four studies (38, 42, 43, 45) had *a high risk of bias*.

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Table 2

Newcastle-Ottawa Scale for case control studies: points awarded to the included studies investigating the association between the auditory system and SIDS.

	Is the case definition adequate [a]	Representativeness of the cases [b]	Selection of controls [c]	Definition of controls [d]	Comparability of cases and controls [e]	Ascertainment of exposure [f]	Same methods of ascertainment for cases and controls [g]	Non-response rate [h]	Risk of bias [i]
Blair 2019 (34)	A (✷)	B	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	C	Low
Rubens 2008 (35)	A (✷)	B	A (✷)	A (✷)	B (✷)	B (✷)	A (✷)	A (✷)	Low
Lavezzi 2015 (36)	A (✷)	A (✷)	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Low
Lavezzi 2015 (37)	A (✷)	A (✷)	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Low
Lavezzi 2012 (38)	C	B	A (✷)	A (✷)	-	A (✷)	A (✷)	A (✷)	High
Oehmichen 1989 (39)	A (✷)	B	A (✷)	A (✷)	A, B (✷✷)	C	A (✷)	A (✷)	Low
Rickert 2004 (40)	A (✷)	A (✷)	A (✷)	A (✷)	B (✷)	C	A (✷)	A (✷)	Low
Gupta 1981 (41)	B	B	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Fair
Kileny 1982 (42)	B	B	C	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	High
Orlowski 1982 (43)	A (✷)	B	C	B	-	A (✷)	A (✷)	A (✷)	High
Pettigrew 1985 (44)	B	B	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Fair
Stockard 1982 (45)	A (✷)	B	C	B	B (✷)	A (✷)	A (✷)	A (✷)	High

[a] A) yes, with independent validation; B) yes, e.g. record linkage or based on self-reports; C) no description. [b] A) consecutive or obviously representative series of cases; B) potential for selection biases or not stated. [c] A) community controls; B) hospital controls; C) no description. [d] A) no history of endpoint; B) no description of source. [e] A) study controls for sleeping position and conditions; B) study controls for at sex, smoking exposure, birth weight or gestational age. [f] A) secure record (e.g. surgical records); B) structured interview where blind to case/control status; C) written self-report or medical record only; D) no description. [g] A) yes; B) no. [h] A) same rate for both groups; B) non-respondents described; C) rate different and no designation. [I] "low risk" was given 3-4 points in selection and 1-2 points in comparability and 2-3 points in outcome; "fair risk" was given 2 points in selection and 1-2 points in comparability and 2-3 points in outcome; "high risk' was given 0-1 point in selection or 0 point in comparability or 0-1 point in outcome.

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Quality of evidence

The included studies were initially rated of low quality due to their observational nature. The quality of evidence across all outcomes was very low. This was mainly due to serious concerns in risk of bias within studies and inconsistency between the study results. Furthermore, NMSIDS as a surrogate outcome for SIDS were downgraded for indirectness.

Risk of bias across studies

Publication bias due to small study effects appears to be unlikely as several studies reported no association between auditory pathology and SIDS or NMSIDS. Selective outcome reporting bias was not detected as all outcomes in the included studies were reported and described in the method sections.

Discussion

We identified 12 case-control studies that were eligible for this systematic review of the association between auditory system pathology and SIDS. Seven studies (35-38, 40, 43, 44) reported that abnormalities in the auditory system measured either by function (hearing screen) or autopsy (histology or immunohistochemistry) were more frequent in SIDS/NMSIDS infants than in controls. Five studies (34, 39, 41, 42, 45) found no association. The results of two studies (44, 45) tended towards an abnormal ABR in some NMSIDS cases with prolonged interpeak intervals indicating neuronal dysfunction. Studies based on histology showed a more pronounced and consistent association than studies of function measures. However, if the three histologic studies by Lavezzi *et al.* (36-38) have overlapping populations the association is less convincing.

This systematic review included three different exposures: OAE, ABR and brainstem histology of the auditory system and two different outcomes: SIDS and NMSIDS. The different exposures and outcomes make it difficult to compare studies and may interfere with the overall conclusion.

Abnormal OAE was present in one out of two studies. Rubens *et al.* (35) found a lower response on the right ear than the left ear. Studies (50-52) have shown that newborn OAE and ABR responses are greater on the right ear compared to the left ear, which is consistent with the control group in the study of Rubens *et al.* (35). Therefore, Rubens *et al.* (35) hypothesized that an inner ear injury may be due to high fetal venous blood pressure during delivery. The high blood pressure may damage the small auricular veins because they drain directly to the jugular veins. However, the left ear may be protected against this high pressure by a greater length and angulation of the left brachiocephalic vein compared

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to the right side.(53, 54) An inner ear lesion may result in a vulnerable infant at risk for SIDS. However, further research is needed to investigate whether this hypothesis may explain SIDS. Examination of OAEs in a larger study of SIDS cases found no abnormalities on the right ear but three had left-sided hearing loss.(55) The study of Chan *et al.* (56) did not report findings in support of the hypothesis by Rubens *et al.* (35). Blair *et al.* (34) found no association between abnormal OAE and SIDS, while still finding an association between SIDS and known risk factors such as co-sleeping, prone sleeping, and being sick. Rubens *et al*. (35) provided no SIDS definition and used a paired *t*-test between cases and controls only matched by gender, date of birth, and neonatal intensive care unit (NICU) versus routine nursery admission, which may devaluate the evidence from the study.(55, 57, 58)

ABR was used in the five NMSIDS studies. These studies had inconsistent results which affect the generalisability.(41-45) Only Orlowski *et al.* (43) found consistent signs of abnormal ABR in NMSIDS infants, while Pettigrew *et al*. (44) found abnormal ABR in 12% of the NMSIDS cases. However, some infants with severe NMSIDS episodes had normal ABRs which imply uncertainty of the results.(44) In contrary to Rubens *et al.* (35), left-sided abnormalities were most frequent in Orlowski *et al.*(43) Three studies found normal ABRs in NMSIDS cases.(41, 42, 45) This inconsistency has also been shown in similar studies with ABR in infants with NMSIDS events.(59-61) Additionally, Brinsmaed *et al.* (62) could not demonstrate abnormal ABR results in SIDS infants. By full-text screening, these studies (59-62) were excluded from this review due to missing information about NMSIDS infants and missing reference group. There may be some concerns regarding the evidence from the NMSIDS studies. Firstly, the studies were assessed by NOS to have a higher degree in risk of bias compared to the SIDS studies. Secondly, the studies on NMSIDS are from the 1980s when the SIDS incidence was higher and the knowledge on SIDS risk factor were less. Therefore, generalisability to current time may be questionable due to different patient population. Thirdly, a NMSIDS event may not be associated with SIDS. (5-7) Actually, the objectives in the NMSIDS studies were to examine ABR as a tool to examine the brainstem generally rather than the auditory function.

Four histological studies (36-38, 40) indicated that abnormalities in the auditory pathway of the brainstem occurred more frequently in SIDS infants. However, comparisons between studies are difficult due to the examination of different structures, cells, and proteins. Though, three studies all found abnormalities in the IC.(36, 37, 40) The expression of c-jun may indicate neuronal injury in the IC.(40, 49) Rickert *et al.* (40) had no mention of whether the histological examination was blinded;

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which may interfere with the interpretation as many of the measures were qualitative rather than quantitative. The three studies by Lavezzi *et al.* (36-38) all stated that the examination of the different nuclei was blinded. The histological studies support that abnormalities in the auditory system may be associated with SIDS. If the histological abnormalities in the auditory system may be captured by a newborn hearing screening, the number of SIDS cases could potentially be reduced. However, the inconsistent ABR and OAE studies do not indicate this as a possibility. Additionally, the confidence intervals of the calculated odds ratios in the three studies of Lavezzi *et al.* (36-38) were wide which may indicate uncertainty of the results.

A challenge in all studies was the missing information on potential confounding factors. Information was limited or absent on factors such as sleeping position, sleeping conditions, maternal smoking, gestational age at birth, birth weight, and sex. For example, maternal smoking during pregnancy increases the risk of hearing disorders and preterm birth and is also a known risk factor for SIDS.(63-65) The lack of adjustment for these factors may have cause an overestimation of the association between auditory system pathology and SIDS. However, due to the small study populations adjustment for multiple factors were impossible.

The risk of bias was lower in studies concerning SIDS compared to NMSIDS. However, all studies had poor comparability between cases and controls which affected the assessment of the quality of evidence. Oehmichen *et al.* (39) was the only study achieving two points in comparability during the NOS assessment; however, the study only adjusted for sleeping position and sex. SIDS is a rare outcome resulting in few SIDS cases. Included studies were underpowered due to small study sizes. Generally, the study results were inconsistent and further studies with larger study populations are needed to investigate the association between the auditory system pathology and SIDS. The quality of evidence across all outcomes was very low, which limits any clear conclusions.

Strengths and limitations of this study

This systematic review was conducted in accordance with PRISMA guidelines with studies searched in different databases. To minimize the overall risk of bias, each step was performed independently by two reviewers. The possibility of publication bias seemed unlikely, but we were unable to assess this further by formal test of asymmetry. Our inclusion criteria were broad to ensure that all relevant studies were included. However, consequently, we included studies with no or inadequate SIDS and NMSIDS definitions increasing the heterogeneity. Furthermore, the comparability between SIDS and NMSIDS

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cases and controls was limited, which may hamper the overall conclusion. At last, the included studies were small resulting in imprecision of estimates.

Conclusion

Despite many years of investigation, it is still impossible to conclude whether auditory system pathology is associated with SIDS. Four out of five histological studies indicate an association between the auditory system and SIDS; particularly abnormalities in the inferior colliculus. However, studies investigating the OAE or ABR and SIDS were inconsistent. The studies included in this review were generally of poor quality and future studies should focus on studying the association between auditory system pathology and SIDS in larger current study populations.

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Availability of data and materials

All data for this study is included in the article and the additional files.

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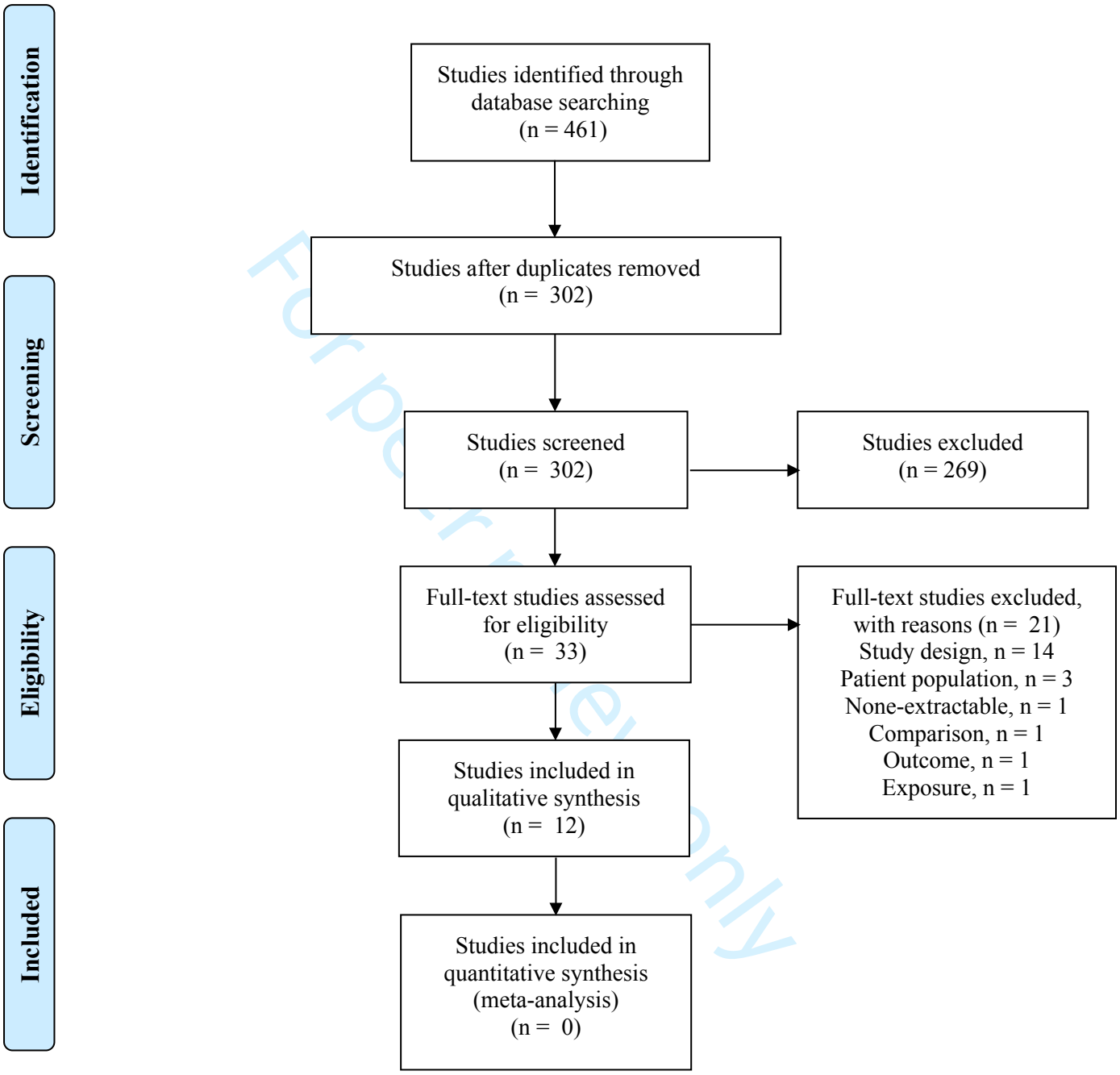
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Figure 1. Prisma flow chart showing the selection process of the systematic review



From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). *P*referred *R*eporting *I*tems for *S*ystematic Reviews and *M*eta-*A*nalyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit www.prisma-statement.org.

Additional file 1

The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

Katrine Dahl, Mads Andersen, Tine B. Henriksen

The full search strategy

Search strategy for Pubmed

1. "Sudden Infant Death" [Mesh]
2. sudden infant death
3. SIDS
4. cot death
5. crib death
6. apparent life threatening event
7. "Infantile Apparent Life-Threatening Event" [Mesh]
8. inner ear
9. auditory
10. "Auditory Pathways" [Mesh]
11. hearing
12. "Hearing" [Mesh]
13. "Hearing tests" [Mesh]
14. cochlear
15. "Cochlear Nucleus" [Mesh]
16. inferior colliculus
17. inferior colliculi
18. "Inferior Colliculi" [Mesh]
19. otoacoustic emission*
20. auditory brainstem response
21. "Evoked Potentials, Auditory, Brain Stem" [Mesh]
22. #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7
23. #8 OR #9 OR #10 OR #11 OR #12 OR #13 #14 #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21
24. #22 AND #23

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Search strategy for Embase

1. "sudden infant death syndrome"/exp
2. sudden infant death
3. SIDS
4. cot death
5. crib death
6. "near sudden infant death syndrome"/exp
7. apparent life threatening event
8. "apparent life threatening event"/exp
9. inner ear
10. "inner ear"/exp
11. auditory
12. "auditory system"/exp
13. hearing
14. "hearing test"/exp
15. cochlear
16. inferior colliculus
17. inferior colliculi
18. otoacoustic emission
19. otoacoustic emissions
20. auditory brainstem response
21. #1 OR #2 OR #3 OR #4 OR #5 OR #6 #7 OR #8
22. #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20
23. #21 AND #22

Search strategy for Web of Science

1. sudden infant death
2. SIDS
3. cot death
4. crib death
5. apparent life threatening event
6. inner ear

For peer review only

7. auditory
8. hearing
9. cochlear
10. inferior colliculus
11. inferior colliculi
12. otoacoustic emission
13. otoacoustic emissions
14. auditory brainstem response
15. #1 OR #2 OR #3 OR #4 OR #5
16. #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14
17. #15 AND #16

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Additional file 2

The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

Katrine Dahl, Mads Andersen, Tine B. Henriksen

Excluded studies with reasons for exclusion

Title	Authors	Reason for exclusion
Is automated auditory brainstem response screening predictive of sudden infant death syndrome?	Brinsmead, T.L; Jardine, L; Thomas, D.	Study design
Neonatal otoacoustic emission screening and sudden infant death syndrome	Chan, R.S.Y.; McPherson, B; Zhang, V.W.	Study design
Inner ear and sudden infant death: Exploring implications for distubances in brainstem functions	Cohen, M.C.; Donn, T; Rivolta, M.; Cacciabue-Rivolta, D; Whitby, E; Rubens, D.	Study design
Brainstem auditory evoked-potentials in near-miss sudden infant death syndrome	Dorfman, L.J.; Gupta, P.R; Guilleminault, C.	Study design
Newborn hearing screen results for infants that died of SIDS in Michigan 2004-2006	Farquhar, L; Jennings, P.	Study design
Sudden infant death syndrome and possible relation to vestibular function	Farrimond, T.	Study design
Otoacoustic emissions does not currently have ability to detect SIDS	Hamill, T.; Lim, G.	Study design
Anatomo-pathologic study of the temporal bone in our 4 cases of sudden infant death	Josset, P.; Lecomte, D.; Omur, M.;	Study design

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	Chabolle, F.; Hannoun, L; Chouard, C.H.	
Auditory arousal thresholds of normal and near-miss SIDS infants	Kahn, A.; Picard, E.; Blum, D.	Outcome
Newborn hearing screens and SIDS	Krous, H.; Byard, R.W.	Study design
Histological study of the middle ear in the sudden infant death syndrome	Lecomte, D.; Josset, P.	Study design
Far-field auditory evoked potentials in near-miss sudden infant death syndrome	Luders, H.; Orlowski, J.P.; Dinner, D.S.	Study design
Abnormal brain stem potentials in infants with threatened sudden infant death syndrome	Nodar, R.H.; Lonsdale, D.; Orlowski, J.P.	Patient population
Abnormal brainstem auditory evoked potentials in infants with threatened sudden infant death syndrome	Orlowski, J.P.; Nodar, R.H.; Lonsdale, D.	Patient population
Prediction of apnea in siblings of sudden infant death syndrome victims using brain-stem auditory evoked-potentials	Orlowski, J.P.; Lonsdale, D.; Nodar, R.H.; Hunt; Strauss; Keenan; Rich.	None-extractable
Brainstem auditory evoked potentials in infants at risk of sudden infant death syndrome (SIDS) with and without aminophylline treatment	Reiterer, F.; Litscher, G.; Kurz, R.; Haidmayer, R.; Pfurtscheller, G.	Exposure
Medullary astrogliosis in sudden infant death syndrome varies with sleeping environment: Evidence for different	Spinelli, J.; Byard, R.W.; Van Den	Comparison

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mechanisms of death in alone versus co-sleepers?	Heuvel, C.; Collins-Praino, L.E.	
Brain-stem auditory evoked-potentials in sudden infant death syndrome (SIDS), near-miss-for-SIDS, and infant apnea syndromes	Stockard, J.J.; Hecox, K.	Study design
Hearing screening and sudden infant death syndrome	Stokowski, L.A.	Study design
Observations of otoconial membranes from human infants	Wright, C.G.; Hubbard, D.G.	Patient population
Absence of otoconia in a human infant	Wright, C.G.; Hubbard, D.G.; Graham, J.W.	Study design

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Reporting checklist for systematic review (with or without a meta-analysis).

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

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		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	1
Abstract			
Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	2
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	4
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	4
Methods			

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Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	5
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	6
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	Additional file 1
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and, if applicable, details of automation tools used in the process	6
Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	6
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	5
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	6
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	6
Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk	n/a

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		ratio, mean difference) used in the synthesis or presentation of results ***The different studies were too heterogeneous to plan any calculation of effect measures and a narrative synthesis was planned.***	
Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	6
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	7
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	7
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	7
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression) ***Not applicable.***	n/a
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results ***Not applicable.***	n/a
Reporting bias assessment	#14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	7
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	7
Results			
Study selection	#16a	Describe the results of the search and selection process, from	8

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		the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	Figure 1
Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	Additional file 2
Study characteristics	#17	Cite each included study and present its characteristics	8-11 incl. table 1
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	13-15 incl. table 2
Results of individual studies	#19	For all outcomes, present for each study (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	Table 1
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	13
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	Table 1
Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	Table 1
Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results ***Not applicable.***	n/a
Risk of reporting biases in syntheses	#21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	16
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	16

Discussion

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Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	16-17
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	18
Limitations of the review methods	#23c	Discuss any limitations of the review processes used	18
Implications	#23d	Discuss implications of the results for practice, policy, and future research	18
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	5
Registration and protocol	#24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	5
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol ***No amendments done.***	n/a
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	3
Competing interests	#26	Declare any competing interests of review authors	3
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	19

None The PRISMA checklist is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist can be completed online using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

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Title page

Title: The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

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Keywords

Sudden Infant Death Syndrome; Hearing screen; Otoacoustic emission; Auditory brainstem response; autopsy; auditory system.

Abstract

Objective A theory has emerged, suggesting that abnormalities in the auditory system may be associated with Sudden Infant Death Syndrome (SIDS). However, the current clinical evidence has never been systematically reviewed.

Design A systematic review was conducted according to the guideline of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).

Data sources PubMed, Embase and Web of Science were systematically searched through 7 September 2020.

Eligibility criteria for selecting studies Only human studies with a reference group were included. Studies were eligible for inclusion if they examined infants exposed to otoacoustic emissions, auditory brainstem response, or had autopsies with brainstem histology of the auditory system. SIDS was the primary outcome, while the secondary outcome was near-miss SIDS episodes (NMSIDS).

Data extraction and synthesis Two independent reviewers extracted data and assessed risk of bias, and the quality of evidence. Due to high heterogeneity, a narrative synthesis was conducted. Risk of bias and quality of evidence was assessed using the Newcastle-Ottawa Scale and Grading of Recommendations Assessment, Development and Evaluation (GRADE).

Results Twelve case-control studies were included. Seven studies on otoacoustic emission (OAE) or auditory brainstem response (ABR) had a high degree of inconsistency. Contrarily, four out of five studies reporting on brainstem histology found that auditory brainstem abnormalities were more prevalent in SIDS cases than in controls. However, the quality of evidence across all studies was very low.

Conclusions This systematic review found no clear association between auditory system pathology and SIDS. The higher prevalence of histological abnormalities in the auditory system of SIDS may indicate an association. However, further studies of higher quality and larger study populations are needed to determine whether these findings are valid.

Funding: No funding received.

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Article Summary

Strength and limitations of this study

1. This is the first systematic review based on human studies that investigates the association between the auditory system pathology and sudden infant death syndrome (SIDS).
2. The systematic review was completed according to the guideline of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).
3. Risk of bias was assessed systematically with the validated Newcastle-Ottawa Scale for case-control studies.
4. Due to the heterogeneity of the included studies, this is a narrative systematic review without meta-analysis.

Abbreviations

SIDS = Sudden infant death syndrome

NMSIDS = Near miss sudden infant death syndrome

ALTE = Apparent life-threatening event

OAE = Otoacoustic emission

ABR = Auditory Brainstem Response

SOC = Superior olivary complex

MSO = Medial superior olivary complex

IC = Inferior Colliculus

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Introduction

Sudden Infant Death Syndrome (SIDS) is defined as the sudden death of an infant under the age of one year occurring unexpectedly, with no other potential explanation of death and an autopsy compatible with SIDS.(1) The recommendation for infants to sleep in a supine position has significantly decreased the incidence of SIDS.(2) In the UK, the SIDS incidence decreased from 2.3 per 1,000 live births in 1988 to 0.77 per 1,000 live births in 1992 after the "back to sleep" campaign with similar trends being observed in other western countries.(3, 4) However, SIDS is still one of the leading causes of death in infancy (0.30 per 1,000 live births) and a scientific mystery.(4) A near-miss SIDS (NMSIDS) episode may be considered as a precursor before an actual SIDS event. However, the evidence for this association is limited.(5-7) NMSIDS is also known as an apparent life-threatening event (ALTE), which to some degree have replaced the term NMSIDS.(7) Previous studies have examined infants surviving NMSIDS episodes anticipating to improve the understanding of SIDS.(8, 9)

Even though SIDS is a diagnosis of exclusion, deficiencies in the cardiorespiratory system and arousal response are believed to play an important role in SIDS.(10-12) Human and animal studies have indicated that the inner ear is connected to the respiratory system and may be essential for the arousal response.(13-18) Furthermore, the anatomical proximity between the respiratory system and the auditory pathway in the brainstem imply that there may be synaptic connections.(19, 20) If abnormalities in the auditory system are associated with SIDS and may be captured by newborn hearing screening, we would be able to follow infants at risk and potentially reduce the number of SIDS cases further. Two methods are used for newborn hearing screenings: Otoacoustic Emission (OAE) (21, 22) and Auditory Brainstem Response (ABR) (23, 24). OAE examines hearing from the external ear to the cochlea, while ABR tests the entire auditory pathway reaching the auditory cortex. Furthermore, if abnormalities of the auditory pathway can be shown histologically, we would get one step closer to a biological understanding of the cause of SIDS.

Several studies have investigated this association. However, the current human evidence has never been compiled. This systematic review aims to provide a comprehensive overview of the literature on the association between auditory system pathology measured by newborn hearing screening or autopsy with histology of the brainstem and Sudden Infant Death Syndrome (SIDS). Additionally, NMSIDS and ALTE are included as secondary outcomes to investigate the possible association.

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Methods

This systematic review was conducted according to the guideline of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).(25) A protocol with predefined methods, search strategy, and inclusion criteria was registered at the International Prospective Register of Systematic Reviews (PROSPERO) 7 September 2020 (reg. number: CRD42020208045).

Study selection and eligibility criteria

Studies, languages, and population

Case-control and cohort studies of infants below one year of age were eligible for inclusion. Experimental studies and studies without a reference group were excluded. We included studies in English, Danish, and other languages with adequate translation by Google Translate. No limitations regarding publication date or location were implemented.

Exposure

Auditory system pathology was defined by abnormal OAE, ABR, or autopsy of the auditory system with histology of the brainstem including the cochlear nuclei, the superior olivary complex (SOC), and the inferior colliculus (IC).(26)

Comparators

In case-control studies, SIDS cases were compared to survivors (ABR or OAE studies) or infants dying from other well-defined causes (autopsy studies). In cohort studies, infants with normal OAE or ABR served as the reference.

Types of outcome

The primary outcome was SIDS defined as the sudden death of a child under the age of one year that occurred unexpectedly and with an autopsy showing no other potential explanation of death.(1) Studies with no or other definitions were still included but rated accordingly by the Newcastle-Ottawa Scale (NOS) for case-control and cohort studies. We included NMSIDS and ALTE as secondary outcomes. NMSIDS and ALTE were defined as severe life-threatening apnoea with marked changes in muscle tone, colour change, choking or gasping, and an apparent need for resuscitation by vigorous stimulation or ventilation.(9, 27)

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Search strategy

The literature search was conducted in Pubmed, Embase, and Web of Science. A search strategy was developed for each database using words related to the auditory system and SIDS. Medical Subject Headings (MeSH) in Pubmed and the Explode function in Embase was used. References of included studies were scrutinized as well as citations identified in Scopus. The search strategy was developed by the reviewers and peer-reviewed by a scientific librarian. The search was performed on 8 September 2020. The complete search strategy is available in Additional file 1. Additionally, we hand-searched grey literature at Open Grey, Ovid, and The National Technical Information Service without any additional results.(28)

Study selection

The search results were transferred to Endnote X9©(29) and duplicates were removed. The search results were then uploaded to Covidence (30) to manage the study selection process. Titles and abstracts were screened and any articles that seemingly met the inclusion criteria were extracted for full-text analysis. This process was conducted independently by two reviewers (KD and MA) with any disagreements resolved by discussion or by a third reviewer (TBH).

Data extraction

Data extraction was performed by two reviewers (KD and MA). The data were extracted using a predefined template, which was tested before use to identify any missing data plots. Authors of eligible studies were not contacted regarding missing information. The following data were extracted from the included studies: 1) title, authors, year of publication, country, study design, and aim of the study; 2) characteristics of the study population including age, sex, gestational age at birth, birth weight, sleeping position and condition, maternal smoking, and socioeconomic status; 3) methods of hearing assessment and autopsy; 4) definition of SIDS, NMSIDS and/or ALTE; 5) point estimates and statistics; and 6) information for assessment of risk of bias and quality of evidence by the Newcastle-Ottawa Scale (NOS) and the Grading of Recommendations Assessment, Development and Evaluation (GRADE).

Risk of bias

A modified version of the NOS for case-control studies and cohort studies was used to assess the risk of bias.(31) The assessment was performed independently by two reviewers (KD and MA) and any disagreements were resolved by discussion or by a third reviewer (TBH). Each study was evaluated

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within three categories; the selection process, the comparability between groups, and the exposure. A maximum of nine points (*) could be awarded to each study by NOS. “Low” risk of bias required 3-4 points in selection, 1-2 points in comparability, and 2-3 points in exposure; “fair” risk required 2 points in selection, 1–2 points in comparability, and 2–3 points in exposure; and ‘high” risk was given 0-1 point in selection or 0 points for comparability or 0-1 point in exposure. Regarding comparability, one point was awarded if the study adjusted or matched for sleeping position or condition, while an additional point was awarded if they adjusted for either maternal smoking, sex, gestational age at birth, or birth weight.

Synthesis

The heterogeneity between included studies was too high to allow for a meta-analysis. A narrative synthesis was therefore performed, comparing differences and similarities of the studies according to Ryan *et al.*(32) Any missing estimates from the studies were calculated by use of GraphPad Prism version 8.00 for MacOS.(33) Statistically significance was defined as a two-sided *p*-value less than 0.05 and a 95% confidence interval (CI) was used.

Quality of evidence

The quality of evidence by GRADE was assessed independently by two reviewers (KD and MA).(34) Five factors could lower the quality: risk of bias, inconsistency, indirectness, imprecision, and publication bias. Serious limitations in any of these five factors resulted in a downgrading of the overall rating. The limitations could either be serious or very serious and downgraded one or two points. Regarding risk of bias, the studies were downgraded if they achieved one or zero points in comparability. Furthermore, they were downgraded one point if they missed a point in the selection process and exposure, or downgraded two points if they miss two points or more in these categories. The overall quality was rated as high, moderate, low, or very low.(33)

Risk of bias across studies

Due to high heterogeneity, a formal test of asymmetry could not be performed. Therefore, a qualitative analysis of publication bias was conducted with consideration of variability in study sample sizes. Furthermore, selective outcome reporting was assessed by comparing outcomes reported in the method and result section of the included studies.

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Patient and public involvement

No patient is involved.

Results

Study selection

A total of 461 studies were identified by the search strategy. After the removal of duplicates, 302 studies were screened by title and abstract. Subsequently, 33 studies were screened by full text and a total of 12 case-control studies were included.(35-46) No cohort studies were identified. No studies in other languages than English were included and no additional studies were identified by snowball search or search for grey literature. The selection process is documented in a PRISMA flowchart in figure 1.(25) Additional file 2 contains an overview of excluded studies with reasons for exclusion.

Study characteristics

Table 1 shows the main characteristics of each study including study period, country, population, outcome definition (SIDS and NMSIDS), methods for the examination of the auditory system, and the provided estimations. The studies included a total of 335 cases and 392 controls. A total of 209 SIDS cases (35-41) and 126 infants with an NMSIDS episode (42-46) were included. OAE or ABR was assessed in 193 cases and 327 controls.(35, 36, 42-46) Histological examination of the brainstem was evaluated in 142 cases and 65 controls.(37-41) Three Italian studies of Lavezzi *et al.* (37-39) may have overlapping populations. These studies examined auditory system pathology of the brainstem in infants who died of SIDS.

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Table 1

Overview of study characteristics from 12 included case-control studies on auditory system pathology and sudden infant death syndrome (SIDS) and near-miss SIDS (NMSIDS)

Study	Inclusion period	Country of study population	Study population - How cases and controls are matched.	SIDS and/or NMSIDS definition	Method	No. cases No. controls	Results and effect estimates
				SIDS studies			
Blair, 2019 (35)	2007-2017	UK	Cases: Request was sent to families to contact The Lullaby Trust.(47) Controls: St. Michael's Hospital, Bristol and Birmingham Women's Hospital. Controls that survived the first year of life. Controls matched on sex.	*SIDS*: Each case was categorised according to the Avon clinicopathological system (48) as an unexplained death.	OAE	SIDS cases: 36 Controls: 177	Non-statically significant difference between cases and controls. *p-values* > 0.05
Rubens, 2008 (36)	1993-2005	USA	Cases: infants born in the state of Rhode Island who died of SIDS. Controls: individually matched controls that survived the first year of life. Controls matched by sex, full term vs. preterm and NICU vs. routine nursery admission.	*SIDS* confirmed by autopsy	OAE	SIDS cases: 31 Controls: 31	Statically significance difference between cases and controls at 2, 3 and 4 kHz on the right ear. *p-values* < 0.05
Lavezzi, 2015 (37)	N/R	Italy	Cases and controls: 69 infants were sent to Research Center and diagnosed according to the application of the guidelines stipulated to the Italian law. Controls: dead of well-defined causes. Controls matched on: sex.	*SIDS*: The sudden unexpected death of an infant <1 year of age, with onset of the fatal episode apparently occurring during sleep, that remains unexplained after a thorough investigation, including performance of a complete autopsy and review of the circumstances of death and the clinical history.(1)	Histology of cochlear nuclei, medial superior olivary nucleus (MSO) and inferior colliculus (IC)	SIDS cases: 49 Controls: 20	Abnormal auditory brainstem nuclei: 47% SIDS cases and 10% ($p = 0.003$) - OR = 8.0 [CI: 1.7;38.1] Cochlear nuclei: 39% cases - 0 controls MSO: 24% cases - 10% controls IC: 41% cases - 0 controls

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Lavezzi, 2015 (38)	N/R	Italy	Cases and controls: 44 infants were sent to the Research Center and diagnosed according to the application of the guidelines stipulated by Italian law. Controls: dead of well-defined causes. Controls matched on: sex.	*SIDS*: no precise definition. Though, the Italian law decrees that, all infants suspected for SIDS who died suddenly in Italian regions within the first year of age must undergo an in-depth anatomo-pathological examination.	Histology of IC and superior olivary complex (SOC). Immuno-histochemistry for serotonergic fibers.	SIDS cases: 29 Controls: 15	IC: 59% cases - 0 controls. $p < 0.05$ SOC: 38% cases - 7% controls. $p < 0.01$ - OR=8.6 [CI: 1.0;74.4] Serotonergic immunoreactivity: reduced or absent in 66% cases and none controls. $p < 0.01$
Lavezzi, 2012 (39)	N/R	Italy	No information Controls: dead of well-defined causes.	No definition	Histology of MSO	SIDS cases: 30 Controls: 12	MSO alterations: 40% cases - 17% controls. - OR=3.3 [CI: 0.6;18.0]
Oehmichen, 1982 (40)	N/R	Germany	No information Controls: dead of well-defined causes. Controls matched on: sex and sleeping position.	*SIDS*: sudden and unexpected death with failure of autopsy, histologic evaluation and chemical-toxicologic analysis to demonstrate a definitive cause of death.	Histologic examination with immuno-histochemistry of IC	SIDS cases: 11 Controls: 11	IC: no statistically significant difference concerning gliosis, neither astrocyte morphology or density. Microglia and lysozyme-positive cells were not observed.
Rickert, 2004 (41)	N/R	Germany	Cases: recruited through the German Study on Sudden Infant Death (GeSID). Controls: Controls: dead of well-defined causes. Controls matched on: sex	*SIDS*: The post-mortem examination, including histology, is by definition, negative with no anatomical or toxicological cause for the death ascertained.	Histologic examination with immuno-histochemistry of IC	SIDS cases: 23 Controls: 7	IC: increased c-jun expression compared with controls. $p = 0.008$.
			NMSIDS studies				
Gupta, 1981 (42)	March 1980 to august 1980	USA	Cases: all infants suspected of NMSIDS were hospitalized on Stanford University Hospital. Controls: recruited in the Palo Alto community and from infants delivered at Stanford University Hospital. Controls matched on: sex	*NMSIDS* defined by strict criteria.(49) Episodes during sleep, in which parents believe that infants almost died. Cyanosis, apnoea and loss of consciousness was observed.	ABR	NMSIDS cases: 9 Controls: 9	9/9 normal BAEP Interpeak latencies, wave latencies, interear latency were for all cases $p > 0.1$.
Kileny, 1982 (43)	N/R	Canada	Cases: infants found in their homes who required immediate resuscitation. Controls: normal control infants Controls matched on: sex and birth weight	*NMSIDS*: Infants who had been apnoeic, blue, or pale in their homes who required immediate resuscitation and for whom no cause could be found for this episode following a detailed hospital investigation.	ABR	NMSIDS cases: 11 Controls: 11	11/11 normal ABR No statistically significant difference observed in interpeak latencies. $p > 0.05$

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Orlowski, 1982 (44)	N/R	USA	No information Controls with similar age distribution.	*NMSIDS* = IAS: infants older than 42 weeks postconception and younger than 12 months of chronological age who have experienced a frightening event at home or in the hospital characterized by prolonged apnoea (>20 sec.) requiring vigorous stimulation or resuscitation and not explained by clinical investigation.	ABR	NMSIDS cases: 36 Controls: 25	36/36 abnormal ABR Statistically significant difference for peak latencies I, III and V, and amplitude III. $p < 0.05$ 42% cases had bilateral ABR abnormalities. 47% cases had left sided ABR abnormalities. 11% cases had right sided ABR abnormalities.
Pettigrew, 1985 (45)	N/R	Australia	Cases: infants admitted to hospital following a period of prolonged apnoea during apparent sleep. Controls: referred for study by verbal contact or had returned to hospital for regular postnatal check-up. Controls matched on: sex and prematurity.	*NMSIDS*: a period of prolonged apnoea during apparent sleep which was associated with pallor or cyanosis and which ceased only after vigorous shaking or mouth to mouth resuscitation.	ABR	NMSIDS cases: 63 Controls: 67	88%: normal ABR. 12% abnormal ABR Difference in distribution of V-IIn interpeak interval. $p < 0.05$. Difference in distribution of V-I interpeak interval. $p > 0.05$.
Stockard, 1982 (46)	N/R	USA	No information Controls: normal infants matched on sex.	*NMSIDS*: episodes of prolonged apnoea during sleep in a previously thriving 3-weeks-to-6-month-old infant which result in generalized cyanosis and flaccidity and required resuscitation.	ABR	NMSIDS cases: 7 Controls: 7	7/7 normal ABR 71% cases had I-V interpeak latency 1.2-2.2 SD above normal mean value.

Table 1: The study characteristics including study period, country, study population, outcome definition for SIDS and NMSIDS, methods for examination of the auditory system, and the provided results. Seven SIDS studies and five NMSIDS studies.
OAE: otoacoustic emission, ABR: auditory brainstem response. N/R: not reported. OR: odds ratio. CI: 95% confidence interval

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SIDS, NMSIDS, and controls

Seven studies (35-41) evaluated SIDS cases. All SIDS diagnosis was confirmed by autopsy in all infants. However, SIDS diagnoses were described differently across studies. Two studies (35, 36) included controls with infants who survived the first year of life, while five studies (37-41) included infants dying of well-defined causes. These well-defined causes were described further in four studies (37, 38, 40, 41) Five studies (42-46) investigated infants under the age of one year with an NMSIDS episode. Similarly, the NMSIDS episodes were described in varying detail; however, all studies defined it as an apnoeic episode that required resuscitation. The majority of NMSIDS studies found no clear predisposing or explainable cause for the NMSIDS episode.(42-44, 46) Stockard *et al.* (46) restricted the age criteria from three weeks to six months of life. Orlowski *et al.* (44) used the term Infant Apnea Syndrome (IAS) for NMSIDS. All NMSIDS cases were age-matched with healthy control infants.

The study results

OAE

OAE was applied as the exposure in two studies.(35, 36) The studies used various measures of OAE and different cut-offs for a failed test. Rubens *et al.* (36) examined the signal to noise ratio (SNR) on both ears and discovered that SIDS cases had lower SNRs on the right ear compared with the left ear, which was reverse for controls. SNRs on the right ear were approximately 4 dB lower at 2, 3, and 4 kHz. An abnormal transient evoked otoacoustic emissions (TEOAE) on the right ear was more prevalent in SIDS cases compared with controls. Due to the result of Rubens *et al.* (36), Blair *et al.* (35) assessed the association between an abnormal OAE on the right ear and SIDS. They found no association between abnormal OAE and SIDS. SNRs were marginally higher rather than lower on the right ear among SIDS cases compared with controls, but not significantly.(35)

Histology

Histological and histochemical examination of the brainstem nuclei was performed in five studies.(37-41) Four studies (37-39, 41) found that abnormalities in the auditory structures of the brainstem occurred more frequently in infants with SIDS compared with infants with a well-defined cause of death. All studies used hematoxylin-eosin staining. The three studies of Lavezzi *et al.* (37-39) performed a histological examination, which was carried out independently by two blinded pathologists. Lavezzi *et al.* 2015 (37) discovered hypoplasia of the cochlear nuclei, decreased number of neurons in the medial superior olivary nucleus (MSO), and cytoarchitectural abnormalities of the

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inferior colliculus (IC) in SIDS cases. Overall, 47% of SIDS cases and 10% of controls had two or more abnormalities in the auditory structures. According to this study (37), the odds of SIDS were 8.0 [CI: 1.7;38.1] times as high in infants with abnormal brainstem nuclei compared to infants with normal brainstem nuclei. Lavezzi *et al.* 2015 (38) and Lavezzi *et al.* 2012 (39) found similar abnormalities of the IC and MSO in SIDS cases compared with infants dying of well-defined causes. Furthermore, immunohistochemistry showed absent or very reduced markers of serotonin in the IC in SIDS infants.(38) Rickert *et al.* (41) investigated the expression of c-jun, an immediate-early-gene, which is part of the response to neuronal injury.(41, 50) The study discovered an increased expression of c-jun in the IC by immunohistochemistry in SIDS cases indicating neuronal injury in the IC.(41, 50) Contrarily, the study of Oehmichen *et al.* (40) found no IC abnormalities in SIDS infants when semi-qualitatively evaluation of gliosis was performed.

ABR

Five studies (42-46) assessed the auditory pathway by ABR. Most infants were tested while asleep, but Orlowski *et al.*(44) and Pettigrew *et al.* (45) also allowed for chloral hydrate sedation and quiet rest. Various details of the ABR method were provided in the different studies. Similar ABR tests were mostly performed; however, tests differed by intensity (from 55 to 90 dB HL), use of masking (46), and electrode placement. Though, all infants within the same study were subjected to the same method. Two studies (42, 43) found no association between abnormal ABR and NMSIDS with all NMSIDS infants having normal ABR. Contrarily, Orlowski *et al.* (44) found that left-sided ABR abnormalities appeared more frequently among NMSIDS cases compared with controls. In the study of Pettigrew *et al.* (45), the majority (88%) of NMSIDS infants had normal ABR but approximately 12% had significant different interpeak intervals compared to controls. Stockard *et al.* (46) found normal ABR in all NMSIDS but found a tendency towards a prolonged V-I interpeak interval.

Risk of bias within studies

Table 2 shows the points (*) awarded by NOS for case-control studies. The seven studies on SIDS (35-41) were classified as having *low risk of bias* with exception of Lavezzi *et al.* 2012 (39), who did not describe the selection process or the comparability between infants who died from SIDS and other well-defined causes. Furthermore, two studies (40, 41) did not state whether their histological examination was blinded. The five studies with NMSIDS cases (42-46) had *higher risk of bias*. Evaluation of exposure in all studies indicated little risk of bias. However, none of the studies adjusted

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for all our pre-defined confounders and two studies (39, 44) failed to control for any of them. Table 1 shows which key variables were included in the matching for each study. In total six studies were assessed to having *low risk of bias* (35-38, 40, 41), while two studies (42, 45) had a *fair risk of bias*, and four studies (39, 43, 44, 46) had *a high risk of bias*.

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Table 2

Newcastle-Ottawa Scale for case-control studies: points awarded to the included studies investigating the association between the auditory system and SIDS.

	Is the case definition adequate [a]	Representativeness of the cases [b]	Selection of controls [c]	Definition of controls [d]	Comparability of cases and controls [e]	Ascertainment of exposure [f]	Same methods of ascertainment for cases and controls [g]	Non-response rate [h]	Risk of bias [i]
Blair 2019 (35)	A (✷)	B	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	C	Low
Rubens 2008 (36)	A (✷)	B	A (✷)	A (✷)	B (✷)	B (✷)	A (✷)	A (✷)	Low
Lavezzi 2015 (37)	A (✷)	A (✷)	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Low
Lavezzi 2015 (38)	A (✷)	A (✷)	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Low
Lavezzi 2012 (39)	C	B	A (✷)	A (✷)	-	A (✷)	A (✷)	A (✷)	High
Oehmichen 1989 (40)	A (✷)	B	A (✷)	A (✷)	A, B (✷✷)	C	A (✷)	A (✷)	Low
Rickert 2004 (41)	A (✷)	A (✷)	A (✷)	A (✷)	B (✷)	C	A (✷)	A (✷)	Low
Gupta 1981 (42)	B	B	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Fair
Kileny 1982 (43)	B	B	C	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	High
Orlowski 1982 (44)	A (✷)	B	C	B	-	A (✷)	A (✷)	A (✷)	High
Pettigrew 1985 (45)	B	B	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Fair
Stockard 1982 (46)	A (✷)	B	C	B	B (✷)	A (✷)	A (✷)	A (✷)	High

[a] A) yes, with independent validation (*); B) yes, e.g. record linkage or based on self-reports; C) no description. [b] A) consecutive or obviously representative series of cases (*); B) potential for selection biases or not stated. [c] A) community controls (*); B) hospital controls; C) no description. [d] A) no history of endpoint (*); B) no description of source. [e] A) study controls for sleeping position and conditions (*); B) study controls for sex, smoking exposure, birth weight or gestational age (*). [f] A) secure record (e.g. surgical records) (*); B) structured interviews were blind to case/control status (*); C) written self-report or medical record only; D) no description. [g] A) yes (*); B) no. [h] A) same rate for both groups (*); B) non-respondents described; C) rate different and no designation. [I] "low risk" was given 3-4 points in selection and 1-2 points in comparability and 2-3 points in outcome; "fair risk" was given 2 points in selection and 1-2 points in comparability and 2-3 points in outcome; "high risk' was given 0-1 point in selection or 0 points in comparability or 0-1 point in outcome.

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Quality of evidence

The included studies were initially rated of low quality due to their observational nature. The quality of evidence across all outcomes was very low. This was mainly due to serious concerns in risk of bias within studies and inconsistency between the study results. Furthermore, NMSIDS as a surrogate outcome for SIDS were downgraded for indirectness according to the GRADE assessment, as it is unclear whether NMSIDS is in fact associated with SIDS.

Risk of bias across studies

Publication bias due to small study effects appears to be unlikely as several studies reported no association between auditory pathology and SIDS or NMSIDS. Selective outcome reporting bias was not detected as all outcomes in the included studies were reported and described in the method sections.

Discussion

Statement of principal findings

We identified 12 case-control studies that were eligible for this systematic review of the association between auditory system pathology and SIDS. Seven studies (36-39, 41, 44, 45) reported that abnormalities in the auditory system measured either by function (hearing screen) or autopsy (histology or immunohistochemistry) were more frequent in SIDS/NMSIDS infants than in controls. Five studies (35, 40, 42, 43, 46) found no association. The results of two studies (45, 46) tended towards an abnormal ABR in some NMSIDS cases with prolonged interpeak intervals indicating neuronal dysfunction. Studies based on histology showed a more pronounced and consistent association than studies of functional measures. However, the three histological studies by Lavezzi *et al.* (37-39) may have overlapping populations, which makes the association less convincing.

Functional measures

Rubens *et al.* (36) found a lower response on the right ear than the left ear. Studies (51-53) have shown that newborn OAE and ABR responses are greater on the right ear compared to the left ear, which is consistent with the control group in the study of Rubens *et al.* (36). Therefore, Rubens *et al.* (36) hypothesized that an inner ear injury may be due to high foetal venous blood pressure during delivery. High blood pressure may damage the small auricular veins because they drain directly to the jugular veins. However, the left ear may be protected against this high pressure by a greater length and angulation of the left brachiocephalic vein compared to the right side.(54, 55) A inner ear lesion may

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result in a vulnerable infant at risk for SIDS. Different studies, both the included study of Blair *et al.* (35) and excluded studies (56, 57), have not been able to find the same association. Examination of OAEs in a larger study of SIDS cases found no abnormalities on the right ear but three had left-sided hearing loss.(56) The study of Chan *et al.* (57) did neither report findings in support of the hypothesis by Rubens *et al.* (36). Unfortunately, Rubens *et al*. (36) provided no SIDS definition and used a paired *t*-test between cases and controls only matched by gender, date of birth, and neonatal intensive care unit (NICU) versus routine nursery admission, which may devaluate the evidence from the study.(56, 58, 59)

The other functional test of the auditory system, ABR, was used in the five older NMSIDS studies (42-46). Only Orlowski *et al.* (44) found consistent signs of abnormal ABR in NMSIDS infants but contrary to Rubens *et al.* (36), left-sided abnormalities were most frequent in Orlowski *et al.*(44) Generally the included studies with ABR as exposure (42-46) were inconsistent as the studies with OAE (35, 36). Additionally, Lüders *et al.* (60) and the newer study of Brinsmaed *et al.* (61) could not demonstrate abnormal ABR results in SIDS infants. By full-text screening, these studies (60, 61) were excluded from this review due to missing information about NMSIDS infants and missing reference groups.

Histological measures

The greatest consistency in the results was seen in the histological studies (37-41). However, comparisons between the studies are difficult due to the examination of different cells and proteins. Though three studies all found abnormalities in the IC.(37, 38, 41) The histological studies support that abnormalities in the auditory system may be associated with SIDS. If the histological abnormalities in the auditory system may be captured by a newborn hearing screening, the number of SIDS cases could potentially be reduced. However, the inconsistent ABR and OAE studies do not indicate this as a possibility. Additionally, the confidence intervals of the calculated odds ratios in the three studies of Lavezzi *et al.* (37-39) were wide which indicate uncertainty of the results.

Rickert *et al.* (41) had no mention of whether the histological examination was blinded; which may interfere with the interpretation as many of the measures were qualitative rather than quantitative. In contrast, the three studies by Lavezzi *et al.* (37-39) all stated that the examination of the different nuclei was blinded.

Overall limitations of the studies

In all studies, information was missing on potential confounding factors. Information was limited or

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absent on factors such as sleeping position, sleeping conditions, maternal smoking, gestational age at birth, birth weight, and sex, which all are associated with SIDS. For example, maternal smoking during pregnancy increases the risk of hearing disorders and preterm birth and is also a known risk factor for SIDS.(62-64) The lack of adjustment for these factors may have caused an overestimation of the association between auditory system pathology and SIDS. However, due to the small study populations adjustment for multiple factors were impossible. All studies had poor comparability between cases and controls which affected the assessment of the quality of evidence. Oehmichen *et al.* (40) was the only study achieving two points in comparability during the NOS assessment; however, the study only adjusted for sleeping position and sex.

Additionally, there may be some concern regarding the evidence from the NMSIDS studies. Firstly, we assessed the studies to have a higher degree of risk of bias compared to the SIDS studies. Secondly, the NMSIDS studies are from the 1980s when the SIDS incidence was higher and the knowledge on SIDS risk factor were less. Therefore, generalisability to current time may be questionable due to different patient populations. Thirdly, a NMSIDS event may not be associated with SIDS.(5-7) Actually, the objectives in the NMSIDS studies were to examine ABR as a tool to examine the brainstem generally rather than the auditory function. By these concerns, the NMSIDS studies were of especially poor quality and did not influence our overall conclusion.

SIDS is a rare outcome resulting in few SIDS cases which led to the inclusion of underpowered studies with small study sizes. Generally, the study results were inconsistent and further studies with larger study populations are needed to investigate the association.

Strengths and weaknesses of the systematic review

According to our knowledge, this is the first systematic review about the association between auditory system pathology and SIDS. This systematic review was conducted following PRISMA guidelines with studies searched in different databases. To minimize the overall risk of bias, each step was performed independently by two reviewers. The possibility of publication bias seemed unlikely, but we were unable to assess this further by formal test of asymmetry. Our inclusion criteria were broad to ensure that all relevant studies were included. We included three different exposures: OAE, ABR and brainstem histology of the auditory system and two different outcomes: SIDS and NMSIDS, which were either rather ill-defined or prone to different definitions between studies. Only one study (37) used the San Diego definition of SIDS (1). The differences in exposure and outcome definition

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comprised comparability between studies and in case of inadequate outcome definition, the quality of the individual study was downgraded during NOS assessment.

We had no expert evaluation of the OAE and ABR measures used in the included studies. Consequently, we may have overrated the quality of these. The comparability between SIDS and NMSIDS cases and controls was limited, which may hamper the overall conclusion. At last, the included studies were small resulting in imprecision of estimates.

Conclusion

Despite many years of investigation, it is still impossible to conclude whether auditory system pathology is associated with SIDS. Four out of five histological studies indicate an association between the auditory system and SIDS; particularly abnormalities in the inferior colliculus. However, studies investigating the OAE or ABR and SIDS were inconsistent. The studies included in this review were generally of poor quality and future studies should focus on studying the association between auditory system pathology and SIDS in larger current study populations.

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Figure 1

Prisma flow chart showing the selection process of the systematic review

Contributorship statement

All authors had substantial contributions to this study. KD, MA, and TBH all contributed to the conception and design of the study. KD and MA screened titles and abstracts, subsequently performed full-text screening and data extraction with any disagreements resolved by discussion or by a third reviewer (TBH). KD drafted the systematic review with all authors (KD, MA, and TBH) revising it critically for important intellectual content prior to the submission. All authors approved the final

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version to be submitted. All authors had access to all data and take responsibility for the integrity of the reported findings.

Competing interests

None declared

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Data sharing statement

All data for this study is included in the article and the additional files.

Ethics statement

Ethical approval was not required as no animals or patients were involved. The systematic review included previously published human studies.

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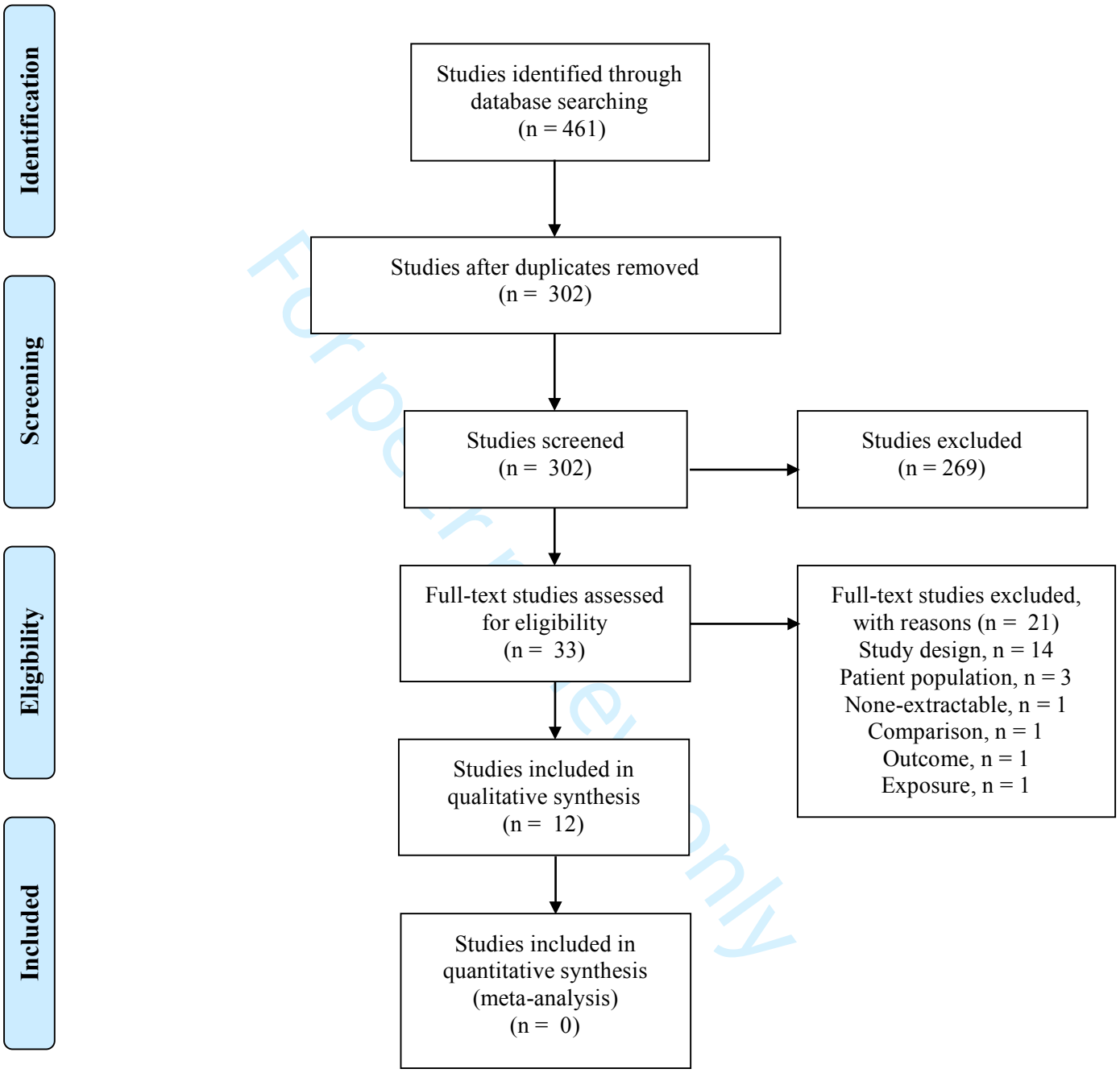
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Figure 1. Prisma flow chart showing the selection process of the systematic review



From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). *P*referred *R*eporting *I*tems for *S*ystematic Reviews and *M*eta-*A*nalyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit www.prisma-statement.org.

Additional file 1

The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

Katrine Dahl, Mads Andersen, Tine B. Henriksen

The full search strategy

Search strategy for Pubmed

1. "Sudden Infant Death" [Mesh]
2. sudden infant death
3. SIDS
4. cot death
5. crib death
6. apparent life threatening event
7. "Infantile Apparent Life-Threatening Event" [Mesh]
8. inner ear
9. auditory
10. "Auditory Pathways" [Mesh]
11. hearing
12. "Hearing" [Mesh]
13. "Hearing tests" [Mesh]
14. cochlear
15. "Cochlear Nucleus" [Mesh]
16. inferior colliculus
17. inferior colliculi
18. "Inferior Colliculi" [Mesh]
19. otoacoustic emission*
20. auditory brainstem response
21. "Evoked Potentials, Auditory, Brain Stem" [Mesh]
22. #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7
23. #8 OR #9 OR #10 OR #11 OR #12 OR #13 #14 #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21
24. #22 AND #23

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Search strategy for Embase

1. "sudden infant death syndrome"/exp
2. sudden infant death
3. SIDS
4. cot death
5. crib death
6. "near sudden infant death syndrome"/exp
7. apparent life threatening event
8. "apparent life threatening event"/exp
9. inner ear
10. "inner ear"/exp
11. auditory
12. "auditory system"/exp
13. hearing
14. "hearing test"/exp
15. cochlear
16. inferior colliculus
17. inferior colliculi
18. otoacoustic emission
19. otoacoustic emissions
20. auditory brainstem response
21. #1 OR #2 OR #3 OR #4 OR #5 OR #6 #7 OR #8
22. #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20
23. #21 AND #22

Search strategy for Web of Science

1. sudden infant death
2. SIDS
3. cot death
4. crib death
5. apparent life threatening event
6. inner ear

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7. auditory
8. hearing
9. cochlear
10. inferior colliculus
11. inferior colliculi
12. otoacoustic emission
13. otoacoustic emissions
14. auditory brainstem response
15. #1 OR #2 OR #3 OR #4 OR #5
16. #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14
17. #15 AND #16

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Additional file 2

The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

Katrine Dahl, Mads Andersen, Tine B. Henriksen

Excluded studies with reasons for exclusion

Title	Authors	Reason for exclusion
Is automated auditory brainstem response screening predictive of sudden infant death syndrome?	Brinsmead, T.L; Jardine, L; Thomas, D.	Study design
Neonatal otoacoustic emission screening and sudden infant death syndrome	Chan, R.S.Y.; McPherson, B; Zhang, V.W.	Study design
Inner ear and sudden infant death: Exploring implications for distubances in brainstem functions	Cohen, M.C.; Donn, T; Rivolta, M.; Cacciabue-Rivolta, D; Whitby, E; Rubens, D.	Study design
Brainstem auditory evoked-potentials in near-miss sudden infant death syndrome	Dorfman, L.J.; Gupta, P.R; Guilleminault, C.	Study design
Newborn hearing screen results for infants that died of SIDS in Michigan 2004-2006	Farquhar, L; Jennings, P.	Study design
Sudden infant death syndrome and possible relation to vestibular function	Farrimond, T.	Study design
Otoacoustic emissions does not currently have ability to detect SIDS	Hamill, T.; Lim, G.	Study design
Anatomo-pathologic study of the temporal bone in our 4 cases of sudden infant death	Josset, P.; Lecomte, D.; Omur, M.;	Study design

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	Chabolle, F.; Hannoun, L; Chouard, C.H.	
Auditory arousal thresholds of normal and near-miss SIDS infants	Kahn, A.; Picard, E.; Blum, D.	Outcome
Newborn hearing screens and SIDS	Krous, H.; Byard, R.W.	Study design
Histological study of the middle ear in the sudden infant death syndrome	Lecomte, D.; Josset, P.	Study design
Far-field auditory evoked potentials in near-miss sudden infant death syndrome	Luders, H.; Orlowski, J.P.; Dinner, D.S.	Study design
Abnormal brain stem potentials in infants with threatened sudden infant death syndrome	Nodar, R.H.; Lonsdale, D.; Orlowski, J.P.	Patient population
Abnormal brainstem auditory evoked potentials in infants with threatened sudden infant death syndrome	Orlowski, J.P.; Nodar, R.H.; Lonsdale, D.	Patient population
Prediction of apnea in siblings of sudden infant death syndrome victims using brain-stem auditory evoked-potentials	Orlowski, J.P.; Lonsdale, D.; Nodar, R.H.; Hunt; Strauss; Keenan; Rich.	None-extractable
Brainstem auditory evoked potentials in infants at risk of sudden infant death syndrome (SIDS) with and without aminophylline treatment	Reiterer, F.; Litscher, G.; Kurz, R.; Haidmayer, R.; Pfurtscheller, G.	Exposure
Medullary astrogliosis in sudden infant death syndrome varies with sleeping environment: Evidence for different	Spinelli, J.; Byard, R.W.; Van Den	Comparison

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mechanisms of death in alone versus co-sleepers?	Heuvel, C.; Collins-Praino, L.E.	
Brain-stem auditory evoked-potentials in sudden infant death syndrome (SIDS), near-miss-for-SIDS, and infant apnea syndromes	Stockard, J.J.; Hecox, K.	Study design
Hearing screening and sudden infant death syndrome	Stokowski, L.A.	Study design
Observations of otoconial membranes from human infants	Wright, C.G.; Hubbard, D.G.	Patient population
Absence of otoconia in a human infant	Wright, C.G.; Hubbard, D.G.; Graham, J.W.	Study design

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Reporting checklist for systematic review (with or without a meta-analysis).

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews

		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	1
Abstract			
Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	2
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	4
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	4
Methods			

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Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	5
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	6
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	Additional file 1
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and, if applicable, details of automation tools used in the process	6
Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	6
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	5
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	6
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used,	6

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		how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	
Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results ***The different studies were too heterogeneous to plan any calculation of effect measures and a narrative synthesis was planned.***	n/a
Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	6
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	7
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	7
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	7
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression) ***Not applicable.***	n/a
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results ***Not applicable.***	n/a
Reporting bias	#14	Describe any methods used to assess risk of bias due	7

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assessment		to missing results in a synthesis (arising from reporting biases)	
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	7
Results			
Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	8 Figure 1
Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	Additional file 2
Study characteristics	#17	Cite each included study and present its characteristics	8-11 incl. table 1
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	13-15 incl. table 2
Results of individual studies	#19	For all outcomes, present for each study (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	Table 1
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	13
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	Table 1
Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	Table 1

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Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results ***Not applicable.***	n/a
Risk of reporting biases in syntheses	#21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	16
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	16
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	16-17
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	18
Limitations of the review methods	#23c	Discuss any limitations of the review processes used	18
Implications	#23d	Discuss implications of the results for practice, policy, and future research	18
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	5
Registration and protocol	#24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	5
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol ***No amendments done.***	n/a
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	3

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Competing interests	#26	Declare any competing interests of review authors	3
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	19

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The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

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Title page

Title: The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

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Keywords

Sudden Infant Death Syndrome; Hearing screen; Otoacoustic emission; Auditory brainstem response; autopsy; auditory system.

Abstract

Objective A theory has emerged, suggesting that abnormalities in the auditory system may be associated with Sudden Infant Death Syndrome (SIDS). However, the current clinical evidence has never been systematically reviewed.

Design A systematic review was conducted according to the guideline of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).

Data sources PubMed, Embase and Web of Science were systematically searched through 7 September 2020.

Eligibility criteria for selecting studies Only human studies with a reference group were included. Studies were eligible for inclusion if they examined infants exposed to otoacoustic emissions, auditory brainstem response, or had autopsies with brainstem histology of the auditory system. SIDS was the primary outcome, while the secondary outcome was near-miss SIDS episodes (NMSIDS).

Data extraction and synthesis Two independent reviewers extracted data and assessed risk of bias, and the quality of evidence. Due to high heterogeneity, a narrative synthesis was conducted. Risk of bias and quality of evidence was assessed using the Newcastle-Ottawa Scale and Grading of Recommendations Assessment, Development and Evaluation (GRADE).

Results Twelve case-control studies were included. Seven studies on otoacoustic emission (OAE) or auditory brainstem response (ABR) had a high degree of inconsistency. Contrarily, four out of five studies reporting on brainstem histology found that auditory brainstem abnormalities were more prevalent in SIDS cases than in controls. However, the quality of evidence across all studies was very low.

Conclusions This systematic review found no clear association between auditory system pathology and SIDS. The higher prevalence of histological abnormalities in the auditory system of SIDS may indicate an association. However, further studies of higher quality and larger study populations are needed to determine whether these findings are valid.

Funding: No funding received.

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Article Summary

Strength and limitations of this study

1. This is the first systematic review based on human studies that investigates the association between the auditory system pathology and sudden infant death syndrome (SIDS).
2. The systematic review was reported according to the guideline of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).
3. Risk of bias was assessed systematically with the validated Newcastle-Ottawa Scale for case-control studies.
4. Due to the heterogeneity of the included studies, this is a narrative systematic review without meta-analysis.

Abbreviations

SIDS = Sudden infant death syndrome

NMSIDS = Near miss sudden infant death syndrome

ALTE = Apparent life-threatening event

OAE = Otoacoustic emission

ABR = Auditory Brainstem Response

SOC = Superior olivary complex

MSO = Medial superior olivary complex

IC = Inferior Colliculus

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Introduction

Sudden Infant Death Syndrome (SIDS) is defined as the sudden death of an infant under the age of one year occurring unexpectedly, with no other potential explanation of death and absence of any other pathological diagnosis at autopsy.(1) The recommendation for infants to sleep in a supine position has significantly decreased the incidence of SIDS.(2) In the UK, the SIDS incidence decreased from 2.3 per 1,000 live births in 1988 to 0.77 per 1,000 live births in 1992 after the "back to sleep" campaign with similar trends being observed in other western countries.(3, 4) However, SIDS is still one of the leading causes of death in infancy (0.30 per 1,000 live births) and a scientific mystery.(4) A near-miss SIDS (NMSIDS) episode may be considered as a precursor before an actual SIDS event. However, the evidence for this association is limited.(5-7) NMSIDS is also known as an apparent life-threatening event (ALTE), which to some degree have replaced the term NMSIDS.(7) Previous studies have examined infants surviving NMSIDS episodes anticipating to improve the understanding of SIDS.(8, 9)

Even though SIDS is a diagnosis of exclusion, deficiencies in the cardiorespiratory system and arousal response are believed to play an important role in SIDS.(10-12) Human and animal studies have indicated that the inner ear is connected to the respiratory system and may be essential for the arousal response.(13-18) Furthermore, the anatomical proximity between the respiratory system and the auditory pathway in the brainstem imply that there may be synaptic connections.(19, 20) If abnormalities in the auditory system are associated with SIDS and may be captured by newborn hearing screening, we would be able to follow infants at risk and potentially reduce the number of SIDS cases further. Two methods are used for newborn hearing screenings: Otoacoustic Emission (OAE) (21, 22) and Auditory Brainstem Response (ABR) (23, 24). OAE examines hearing from the external ear to the cochlea, while ABR tests the entire auditory pathway reaching the auditory cortex. Furthermore, if abnormalities of the auditory pathway can be shown histologically, we would get one step closer to a biological understanding of the cause of SIDS.

Several studies have investigated this association. However, the current human evidence has never been compiled. This systematic review aims to provide a comprehensive overview of the literature on the association between auditory system pathology measured by newborn hearing screening or autopsy with histology of the brainstem and Sudden Infant Death Syndrome (SIDS). Additionally, NMSIDS and ALTE are included as secondary outcomes to investigate the possible association.

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Methods

This systematic review was conducted according to the guideline of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).(25) A protocol with predefined methods, search strategy, and inclusion criteria was registered at the International Prospective Register of Systematic Reviews (PROSPERO) 7 September 2020 (reg. number: CRD42020208045).

Study selection and eligibility criteria

Studies, languages, and population

Case-control and cohort studies of infants below one year of age were eligible for inclusion. Experimental studies and studies without a reference group were excluded. We included studies in English, Danish, and other languages with adequate translation by Google Translate. No limitations regarding publication date or location were implemented.

Exposure

Auditory system pathology was defined by abnormal OAE, ABR, or autopsy of the auditory system with histology of the brainstem including the cochlear nuclei, the superior olivary complex (SOC), and the inferior colliculus (IC).(26)

Comparators

In case-control studies, SIDS cases were compared to survivors (ABR or OAE studies) or infants dying from other well-defined causes (autopsy studies). In cohort studies, infants with normal OAE or ABR served as the reference.

Types of outcome

The primary outcome was SIDS defined as the sudden death of a child under the age of one year that occurred unexpectedly and with an autopsy showing no other potential explanation of death.(1) Studies with no or other definitions were still included but rated accordingly by the Newcastle-Ottawa Scale (NOS) for case-control and cohort studies. We included NMSIDS and ALTE as secondary outcomes. NMSIDS and ALTE were defined as severe life-threatening apnoea with marked changes in muscle tone, colour change, choking or gasping, and an apparent need for resuscitation by vigorous stimulation or ventilation.(9, 27)

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Search strategy

The literature search was conducted in Pubmed, Embase, and Web of Science. A search strategy was developed for each database using words related to the auditory system and SIDS. Medical Subject Headings (MeSH) in Pubmed and the Explode function in Embase was used. References of included studies were scrutinized as well as citations identified in Scopus. The search strategy was developed by the reviewers and peer-reviewed by a scientific librarian. The search was performed on 8 September 2020. The complete search strategy is available in Additional file 1. Additionally, we hand-searched grey literature at Open Grey, Ovid, and The National Technical Information Service without any additional results.(28)

Study selection

The search results were transferred to Endnote X9©(29) and duplicates were removed. The search results were then uploaded to Covidence (30) to manage the study selection process. Titles and abstracts were screened and any articles that seemingly met the inclusion criteria were extracted for full-text analysis. This process was conducted independently by two reviewers (KD and MA) with any disagreements resolved by discussion or by a third reviewer (TBH).

Data extraction

Data extraction was performed by two reviewers (KD and MA). The data were extracted using a predefined template, which was tested before use to identify any missing data plots. Authors of eligible studies were not contacted regarding missing information. The following data were extracted from the included studies: 1) title, authors, year of publication, country, study design, and aim of the study; 2) characteristics of the study population including age, sex, gestational age at birth, birth weight, sleeping position and condition, maternal smoking, and socioeconomic status; 3) methods of hearing assessment and autopsy; 4) definition of SIDS, NMSIDS and/or ALTE; 5) point estimates and statistics; and 6) information for assessment of risk of bias and quality of evidence by the Newcastle-Ottawa Scale (NOS) and the Grading of Recommendations Assessment, Development and Evaluation (GRADE).

Risk of bias

A modified version of the NOS for case-control studies and cohort studies was used to assess the risk of bias.(31) The assessment was performed independently by two reviewers (KD and MA) and any disagreements were resolved by discussion or by a third reviewer (TBH). Each study was evaluated

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within three categories; the selection process, the comparability between groups, and the exposure. A maximum of nine points (*) could be awarded to each study by NOS. “Low” risk of bias required 3-4 points in selection, 1-2 points in comparability, and 2-3 points in exposure; “fair” risk required 2 points in selection, 1–2 points in comparability, and 2–3 points in exposure; and ‘high” risk was given 0-1 point in selection or 0 points for comparability or 0-1 point in exposure. Regarding comparability, one point was awarded if the study adjusted or matched for sleeping position or condition, while an additional point was awarded if they adjusted for either maternal smoking, sex, gestational age at birth, or birth weight.

Synthesis

The heterogeneity between included studies was too high to allow for a meta-analysis. A narrative synthesis was therefore performed, comparing differences and similarities of the studies according to Ryan *et al.*(32) Any missing estimates from the studies were calculated by use of GraphPad Prism version 8.00 for MacOS.(33) Statistically significance was defined as a two-sided *p*-value less than 0.05 and a 95% confidence interval (CI) was used.

Quality of evidence

The quality of evidence by GRADE was assessed independently by two reviewers (KD and MA).(34) Five factors could lower the quality: risk of bias, inconsistency, indirectness, imprecision, and publication bias. Serious limitations in any of these five factors resulted in a downgrading of the overall rating. The limitations could either be serious or very serious and downgraded one or two points. Regarding risk of bias, the studies were downgraded if they achieved one or zero points in comparability. Furthermore, they were downgraded one point if they missed a point in the selection process and exposure, or downgraded two points if they miss two points or more in these categories. The overall quality was rated as high, moderate, low, or very low.(33)

Risk of bias across studies

Due to high heterogeneity, a formal test of asymmetry could not be performed. Therefore, a qualitative analysis of publication bias was conducted with consideration of variability in study sample sizes. Furthermore, selective outcome reporting was assessed by comparing outcomes reported in the method and result section of the included studies.

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Patient and public involvement

No patient is involved.

Results

Study selection

A total of 461 studies were identified by the search strategy. After the removal of duplicates, 302 studies were screened by title and abstract. Subsequently, 33 studies were screened by full text and a total of 12 case-control studies were included.(35-46) No cohort studies were identified. No studies in other languages than English were included and no additional studies were identified by snowball search or search for grey literature. The selection process is documented in a PRISMA flowchart in figure 1.(25) Additional file 2 contains an overview of excluded studies with reasons for exclusion.

Study characteristics

Table 1 shows the main characteristics of each study including study period, country, population, outcome definition (SIDS and NMSIDS), methods for the examination of the auditory system, and the provided estimations. The studies included a total of 335 cases and 392 controls. A total of 209 SIDS cases (35-41) and 126 infants with an NMSIDS episode (42-46) were included. OAE or ABR was assessed in 193 cases and 327 controls.(35, 36, 42-46) Histological examination of the brainstem was evaluated in 142 cases and 65 controls.(37-41) Three Italian studies of Lavezzi *et al.* (37-39) may have overlapping populations. These studies examined auditory system pathology of the brainstem in infants who died of SIDS.

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Table 1

Overview of study characteristics from 12 included case-control studies on auditory system pathology and sudden infant death syndrome (SIDS) and near-miss SIDS (NMSIDS)

Study	Inclusion period	Country of study population	Study population - How cases and controls are matched.	SIDS and/or NMSIDS definition	Method	No. cases No. controls	Results and effect estimates
				SIDS studies			
Blair, 2019 (35)	2007-2017	UK	Cases: Request was sent to families to contact The Lullaby Trust.(47) Controls: St. Michael's Hospital, Bristol and Birmingham Women's Hospital. Controls that survived the first year of life. Controls matched on sex.	*SIDS*: Each case was categorised according to the Avon clinicopathological system (48) as an unexplained death.	OAE	SIDS cases: 36 Controls: 177	Non-statically significant difference between cases and controls. *p-values* > 0.05
Rubens, 2008 (36)	1993-2005	USA	Cases: infants born in the State of Rhode Island who died of SIDS. Controls: individually matched controls that survived the first year of life. Controls matched by sex, full term vs. preterm and NICU vs. routine nursery admission.	*SIDS* confirmed by autopsy reports.	OAE	SIDS cases: 31 Controls: 31	Statically significance difference between cases and controls at 2, 3 and 4 kHz on the right ear. *p-values* < 0.05
Lavezzi, 2015 (37)	N/R	Italy	Cases and controls: 69 infants were sent to Research Center and diagnosed according to the application of the guidelines stipulated to the Italian law. Controls: dead of well-defined causes. Controls matched on: sex.	*SIDS*: The sudden unexpected death of an infant <1 year of age, with onset of the fatal episode apparently occurring during sleep, that remains unexplained after a thorough investigation, including performance of a complete autopsy and review of the circumstances of death and the clinical history.(1)	Histology of cochlear nuclei, medial superior olivary nucleus (MSO) and inferior colliculus (IC)	SIDS cases: 49 Controls: 20	Abnormal auditory brainstem nuclei: 47% SIDS cases and 10% controls ($p = 0.003$) OR = 8.0 [CI: 1.7;38.1] Cochlear nuclei abnormalities: 39% cases - 0 controls MSO abnormalities: 24% cases - 10% controls IC abnormalities: 41% cases - 0 controls
Lavezzi, 2015 (38)	N/R	Italy	Cases and controls: 44 infants were sent to the Research Center and diagnosed according to the application of the guidelines stipulated by Italian law. Controls: dead of well-defined causes. Controls matched on: sex.	*SIDS*: no precise definition. Though, the Italian law decrees that, all infants suspected for SIDS who died suddenly in Italian regions within the first year of age must undergo an in-depth anatomo-pathological examination.	Histology of IC and superior olivary complex (SOC). Immuno-histochemistry for serotonergic fibers.	SIDS cases: 29 Controls: 15	IC abnormalities: 59% cases - 0 controls. $p < 0.05$ SOC abnormalities: 38% cases - 7% controls. $p < 0.01$ - OR=8.6 [CI: 1.0;74.4] Serotonergic immunoreactivity: reduced or absent in 66% cases and none controls. $p < 0.01$
Lavezzi, 2012 (39)	N/R	Italy	No information Controls: dead of well-defined causes.	No definition	Histology of MSO	SIDS cases: 30 Controls: 12	MSO abnormalities: 40% cases - 17% controls. - OR=3.3 [CI: 0.6;18.0]
Oehmichen, 1982 (40)	N/R	Germany	No information Controls: dead of well-defined causes. Controls matched on: sex and sleeping position.	*SIDS*: sudden and unexpected death with failure of autopsy, histologic evaluation and chemical-toxicologic analysis to demonstrate a definitive cause of death.	Histologic examination with immuno-histochemistry of IC	SIDS cases: 11 Controls: 11	IC: no statistically significant difference concerning gliosis, neither astrocyte morphology or density. Microglia and lysozyme-positive cells were not observed.

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Rickert, 2004 (41)	N/R	Germany	Cases: recruited through the German Study on Sudden Infant Death (GeSID). Controls: Controls: dead of well-defined causes. Controls matched on: sex	*SIDS*: The post-mortem examination, including histology, is by definition, negative with no anatomical or toxicological cause for the death ascertained.	Histologic examination with immuno-histochemistry of IC	SIDS cases: 23 Controls: 7	IC: increased c-jun expression compared with controls. $p = 0.008$.
NMSIDS studies							
Gupta, 1981 (42)	March 1980 to august 1980	USA	Cases: all infants suspected of NMSIDS were hospitalized on Stanford University Hospital. Controls: recruited in the Palo Alto community and from infants delivered at Stanford University Hospital. Controls matched on: sex	*NMSIDS* defined by strict criteria.(49) Episodes during sleep, in which parents believe that infants almost died. Cyanosis, apnoea and loss of consciousness was observed.	ABR	NMSIDS cases: 9 Controls: 9	9/9 normal ABR Interpeak latencies, wave latencies, interear latency were for all cases $p > 0.1$.
Kileny, 1982 (43)	N/R	Canada	Cases: infants found in their homes who required immediate resuscitation. Controls: normal control infants Controls matched on: sex and birth weight	*NMSIDS*: Infants who had been apnoeic, blue, or pale in their homes who required immediate resuscitation and for whom no cause could be found for this episode following a detailed hospital investigation.	ABR	NMSIDS cases: 11 Controls: 11	11/11 normal ABR No statistically significant difference observed in interpeak latencies. $p > 0.05$
Orlowski, 1982 (44)	N/R	USA	No information Controls with similar age distribution.	*NMSIDS* = IAS: infants older than 42 weeks postconception and younger than 12 months of chronological age who have experienced a frightening event at home or in the hospital characterized by prolonged apnoea (>20 sec.) requiring vigorous stimulation or resuscitation and not explained by clinical investigation.	ABR	NMSIDS cases: 36 Controls: 25	36/36 abnormal ABR Statistically significant difference for peak latencies I, III and V, and amplitude III. $p < 0.05$ 42% cases had bilateral ABR abnormalities. 47% cases had left sided ABR abnormalities. 11% cases had right sided ABR abnormalities.
Pettigrew, 1985 (45)	N/R	Australia	Cases: infants admitted to hospital following a period of prolonged apnoea during apparent sleep. Controls: referred for study by verbal contact or had returned to hospital for regular postnatal check-up. Controls matched on: sex and prematurity.	*NMSIDS*: a period of prolonged apnoea during apparent sleep which was associated with pallor or cyanosis and which ceased only after vigorous shaking or mouth to mouth resuscitation.	ABR	NMSIDS cases: 63 Controls: 67	88%: normal ABR. 12% abnormal ABR Difference in distribution of V-IIn interpeak interval. $p < 0.05$. Difference in distribution of V-I interpeak interval. $p > 0.05$.
Stockard, 1982 (46)	N/R	USA	No information Controls: normal infants matched on sex.	*NMSIDS*: episodes of prolonged apnoea during sleep in a previously thriving 3-weeks-to-6-month-old infant which result in generalized cyanosis and flaccidity and required resuscitation.	ABR	NMSIDS cases: 7 Controls: 7	7/7 normal ABR 71% cases had I-V interpeak latency 1.2-2.2 SD above normal mean value.

Table 1: The study characteristics including study period, country, study population, outcome definition for SIDS and NMSIDS, methods for examination of the auditory system, and the provided results. Seven SIDS studies and five NMSIDS studies. OAE: otoacoustic emission, ABR: auditory brainstem response. N/R: not reported. OR: odds ratio. CI: 95% confidence interval.

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SIDS, NMSIDS, and controls

Seven studies (35-41) evaluated SIDS cases. All SIDS diagnoses were given following the absence of any other pathological diagnosis at autopsy. However, SIDS diagnoses were described differently across studies. Two studies (35, 36) included controls with infants who survived the first year of life, while five studies (37-41) included infants dying of well-defined causes. These well-defined causes were described further in four studies (37, 38, 40, 41) Five studies (42-46) investigated infants under the age of one year with an NMSIDS episode. Similarly, the NMSIDS episodes were described in varying detail; however, all studies defined it as an apnoeic episode that required resuscitation. The majority of NMSIDS studies found no clear predisposing or explainable cause for the NMSIDS episode.(42-44, 46) Stockard *et al.* (46) restricted the age criteria from three weeks to six months of life. Orlowski *et al.* (44) used the term Infant Apnea Syndrome (IAS) for NMSIDS. All NMSIDS cases were age-matched with healthy control infants.

The study results

OAE

OAE was applied as the exposure in two studies.(35, 36) The studies used various measures of OAE and different cut-offs for a failed test. Rubens *et al.* (36) examined the signal to noise ratio (SNR) on both ears and discovered that SIDS cases had lower SNRs on the right ear compared with the left ear, which was reverse for controls. SNRs on the right ear were approximately 4 dB lower at 2, 3, and 4 kHz. An abnormal transient evoked otoacoustic emissions (TEOAE) on the right ear was more prevalent in SIDS cases compared with controls. Due to the result of Rubens *et al.* (36), Blair *et al.* (35) assessed the association between an abnormal OAE on the right ear and SIDS. They found no association between abnormal OAE and SIDS. SNRs were marginally higher rather than lower on the right ear among SIDS cases compared with controls, but not significantly.(35)

Histology

Histological and histochemical examination of the brainstem nuclei was performed in five studies.(37-41) Four studies (37-39, 41) found that abnormalities in the auditory structures of the brainstem occurred more frequently in infants with SIDS compared with infants with a well-defined cause of death. All studies used hematoxylin-eosin staining. The three studies of Lavezzi *et al.* (37-39) performed a histological examination, which was carried out independently by two blinded pathologists. Lavezzi *et al.* 2015 (37) discovered hypoplasia of the cochlear nuclei, decreased number

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of neurons in the medial superior olivary nucleus (MSO), and cytoarchitectural abnormalities of the inferior colliculus (IC) in SIDS cases. Overall, 47% of SIDS cases and 10% of controls had two or more abnormalities in the auditory structures. According to this study (37), the odds of SIDS were 8.0 [CI: 1.7;38.1] times as high in infants with abnormal brainstem nuclei compared to infants with normal brainstem nuclei. Lavezzi *et al.* 2015 (38) and Lavezzi *et al.* 2012 (39) found similar abnormalities of the IC and MSO in SIDS cases compared with infants dying of well-defined causes. Furthermore, immunohistochemistry showed absent or very reduced markers of serotonin in the IC in SIDS infants.(38) Rickert *et al.* (41) investigated the expression of c-jun, an immediate-early-gene, which is part of the response to neuronal injury.(41, 50) The study discovered an increased expression of c-jun in the IC by immunohistochemistry in SIDS cases indicating neuronal injury in the IC.(41, 50) Contrarily, the study of Oehmichen *et al.* (40) found no IC abnormalities in SIDS infants when semi-qualitatively evaluation of gliosis was performed.

ABR

Five studies (42-46) assessed the auditory pathway by ABR. Most infants were tested while asleep, but Orlowski *et al.*(44) and Pettigrew *et al.* (45) also allowed for chloral hydrate sedation and quiet rest. Various details of the ABR method were provided in the different studies. Similar ABR tests were mostly performed; however, tests differed by intensity (from 55 to 90 dB HL), use of masking (46), and electrode placement. Though, all infants within the same study were subjected to the same method. Two studies (42, 43) found no association between abnormal ABR and NMSIDS with all NMSIDS infants having normal ABR. Contrarily, Orlowski *et al.* (44) found that left-sided ABR abnormalities appeared more frequently among NMSIDS cases compared with controls. In the study of Pettigrew *et al.* (45), the majority (88%) of NMSIDS infants had normal ABR but approximately 12% had significant different interpeak intervals compared to controls. Stockard *et al.* (46) found normal ABR in all NMSIDS but found a tendency towards a prolonged V-I interpeak interval.

Risk of bias within studies

Table 2 shows the points (*) awarded by NOS for case-control studies. The seven studies on SIDS (35-41) were classified as having *low risk of bias* with exception of Lavezzi *et al.* 2012 (39), who did not describe the selection process or the comparability between infants who died from SIDS and other well-defined causes. Furthermore, two studies (40, 41) did not state whether their histological examination was blinded. The five studies with NMSIDS cases (42-46) had *higher risk of bias*.

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Evaluation of exposure in all studies indicated little risk of bias. However, none of the studies adjusted for all our pre-defined confounders and two studies (39, 44) failed to control for any of them. Table 1 shows which key variables were included in the matching for each study. In total six studies were assessed to having *low risk of bias* (35-38, 40, 41), while two studies (42, 45) had a *fair risk of bias*, and four studies (39, 43, 44, 46) had *a high risk of bias*.

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Table 2

Newcastle-Ottawa Scale for case-control studies: points awarded to the included studies investigating the association between the auditory system and SIDS.

	Is the case definition adequate [a]	Representativeness of the cases [b]	Selection of controls [c]	Definition of controls [d]	Comparability of cases and controls [e]	Ascertainment of exposure [f]	Same methods of ascertainment for cases and controls [g]	Non-response rate [h]	Risk of bias [i]
Blair 2019 (35)	A (✷)	B	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	C	Low
Rubens 2008 (36)	A (✷)	B	A (✷)	A (✷)	B (✷)	B (✷)	A (✷)	A (✷)	Low
Lavezzi 2015 (37)	A (✷)	A (✷)	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Low
Lavezzi 2015 (38)	A (✷)	A (✷)	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Low
Lavezzi 2012 (39)	C	B	A (✷)	A (✷)	-	A (✷)	A (✷)	A (✷)	High
Oehmichen 1989 (40)	A (✷)	B	A (✷)	A (✷)	A, B (✷✷)	C	A (✷)	A (✷)	Low
Rickert 2004 (41)	A (✷)	A (✷)	A (✷)	A (✷)	B (✷)	C	A (✷)	A (✷)	Low
Gupta 1981 (42)	B	B	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Fair
Kileny 1982 (43)	B	B	C	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	High
Orlowski 1982 (44)	A (✷)	B	C	B	-	A (✷)	A (✷)	A (✷)	High
Pettigrew 1985 (45)	B	B	A (✷)	A (✷)	B (✷)	A (✷)	A (✷)	A (✷)	Fair
Stockard 1982 (46)	A (✷)	B	C	B	B (✷)	A (✷)	A (✷)	A (✷)	High

[a] A) yes, with independent validation (*); B) yes, e.g. record linkage or based on self-reports; C) no description. [b] A) consecutive or obviously representative series of cases (*); B) potential for selection biases or not stated. [c] A) community controls (*); B) hospital controls; C) no description. [d] A) no history of endpoint (*); B) no description of source. [e] A) study controls for sleeping position and conditions (*); B) study controls for sex, smoking exposure, birth weight or gestational age (*). [f] A) secure record (e.g. surgical records) (*); B) structured interviews were blind to case/control status (*); C) written self-report or medical record only; D) no description. [g] A) yes (*); B) no. [h] A) same rate for both groups (*); B) non-respondents described; C) rate different and no designation. [I] "low risk" was given 3-4 points in selection and 1-2 points in comparability and 2-3 points in outcome; "fair risk" was given 2 points in selection and 1-2 points in comparability and 2-3 points in outcome; "high risk' was given 0-1 point in selection or 0 points in comparability or 0-1 point in outcome.

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Quality of evidence

The included studies were initially rated of low quality due to their observational nature. The quality of evidence across all outcomes was very low. This was mainly due to serious concerns in risk of bias within studies and inconsistency between the study results. Furthermore, NMSIDS as a surrogate outcome for SIDS were downgraded for indirectness according to the GRADE assessment, as it is unclear whether NMSIDS is in fact associated with SIDS.

Risk of bias across studies

Publication bias due to small study effects appears to be unlikely as several studies reported no association between auditory pathology and SIDS or NMSIDS. Selective outcome reporting bias was not detected as all outcomes in the included studies were reported and described in the method sections.

Discussion

Statement of principal findings

We identified 12 case-control studies that were eligible for this systematic review of the association between auditory system pathology and SIDS. Seven studies (36-39, 41, 44, 45) reported that abnormalities in the auditory system measured either by function (hearing screen) or autopsy (histology or immunohistochemistry) were more frequent in SIDS/NMSIDS infants than in controls. Five studies (35, 40, 42, 43, 46) found no association. The results of two studies (45, 46) tended towards an abnormal ABR in some NMSIDS cases with prolonged interpeak intervals indicating neuronal dysfunction. Studies based on histology showed a more pronounced and consistent association than studies of functional measures. However, the three histological studies by Lavezzi *et al.* (37-39) may have overlapping populations, which makes the association less convincing.

Functional measures

Rubens *et al.* (36) found a lower response on the right ear than the left ear. Studies (51-53) have shown that newborn OAE and ABR responses are greater on the right ear compared to the left ear, which is consistent with the control group in the study of Rubens *et al.* (36). Therefore, Rubens *et al.* (36) hypothesized that an inner ear injury may be due to high foetal venous blood pressure during delivery. High blood pressure may damage the small auricular veins because they drain directly to the jugular veins. However, the left ear may be protected against this high pressure by a greater length and angulation of the left brachiocephalic vein compared to the right side.(54, 55) A inner ear lesion may

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result in a vulnerable infant at risk for SIDS. Different studies, both the included study of Blair *et al.* (35) and excluded studies (56, 57), have not been able to find the same association. Examination of OAEs in a larger study of SIDS cases found no abnormalities on the right ear but three had left-sided hearing loss.(56) The study of Chan *et al.* (57) did neither report findings in support of the hypothesis by Rubens *et al.* (36). Unfortunately, Rubens *et al*. (36) provided no SIDS definition and used a paired *t*-test between cases and controls only matched by gender, date of birth, and neonatal intensive care unit (NICU) versus routine nursery admission, which may devaluate the evidence from the study.(56, 58, 59)

The other functional test of the auditory system, ABR, was used in the five older NMSIDS studies (42-46). Only Orlowski *et al.* (44) found consistent signs of abnormal ABR in NMSIDS infants but contrary to Rubens *et al.* (36), left-sided abnormalities were most frequent in Orlowski *et al.*(44) Generally the included studies with ABR as exposure (42-46) were inconsistent as the studies with OAE (35, 36). Additionally, Lüders *et al.* (60) and the newer study of Brinsmaed *et al.* (61) could not demonstrate abnormal ABR results in SIDS infants. By full-text screening, these studies (60, 61) were excluded from this review due to missing information about NMSIDS infants and missing reference groups.

Histological measures

The greatest consistency in the results was seen in the histological studies (37-41). However, comparisons between the studies are difficult due to the examination of different cells and proteins. Though three studies all found abnormalities in the IC.(37, 38, 41) The histological studies support that abnormalities in the auditory system may be associated with SIDS. If the histological abnormalities in the auditory system may be captured by a newborn hearing screening, the number of SIDS cases could potentially be reduced. However, the inconsistent ABR and OAE studies do not indicate this as a possibility. Additionally, the confidence intervals of the calculated odds ratios in the three studies of Lavezzi *et al.* (37-39) were wide which indicate uncertainty of the results.

Rickert *et al.* (41) had no mention of whether the histological examination was blinded; which may interfere with the interpretation as many of the measures were qualitative rather than quantitative. In contrast, the three studies by Lavezzi *et al.* (37-39) all stated that the examination of the different nuclei was blinded.

Overall limitations of the studies

In all studies, information was missing on potential confounding factors. Information was limited or

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absent on factors such as sleeping position, sleeping conditions, maternal smoking, gestational age at birth, birth weight, and sex, which all are associated with SIDS. For example, maternal smoking during pregnancy increases the risk of hearing disorders and preterm birth and is also a known risk factor for SIDS.(62-64) The lack of adjustment for these factors may have caused an overestimation of the association between auditory system pathology and SIDS. However, due to the small study populations adjustment for multiple factors were impossible. All studies had poor comparability between cases and controls which affected the assessment of the quality of evidence. Oehmichen *et al.* (40) was the only study achieving two points in comparability during the NOS assessment; however, the study only adjusted for sleeping position and sex.

Additionally, there may be some concern regarding the evidence from the NMSIDS studies. Firstly, we assessed the studies to have a higher degree of risk of bias compared to the SIDS studies. Secondly, the NMSIDS studies are from the 1980s when the SIDS incidence was higher and the knowledge on SIDS risk factor were less. Therefore, generalisability to current time may be questionable due to different patient populations. Thirdly, a NMSIDS event may not be associated with SIDS.(5-7) Actually, the objectives in the NMSIDS studies were to examine ABR as a tool to examine the brainstem generally rather than the auditory function. By these concerns, the NMSIDS studies were of especially poor quality and did not influence our overall conclusion.

SIDS is a rare outcome resulting in few SIDS cases which led to the inclusion of underpowered studies with small study sizes. Generally, the study results were inconsistent and further studies with larger study populations are needed to investigate the association.

Strengths and weaknesses of the systematic review

According to our knowledge, this is the first systematic review about the association between auditory system pathology and SIDS. This systematic review was conducted following PRISMA guidelines with studies searched in different databases. To minimize the overall risk of bias, each step was performed independently by two reviewers. The possibility of publication bias seemed unlikely, but we were unable to assess this further by formal test of asymmetry. Our inclusion criteria were broad to ensure that all relevant studies were included. We included three different exposures: OAE, ABR and brainstem histology of the auditory system and two different outcomes: SIDS and NMSIDS, which were either rather ill-defined or prone to different definitions between studies. Only one study (37) used the San Diego definition of SIDS (1). The differences in exposure and outcome definition

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comprised comparability between studies and in case of inadequate outcome definition, the quality of the individual study was downgraded during NOS assessment.

We had no expert evaluation of the OAE and ABR measures used in the included studies. Consequently, we may have overrated the quality of these. The comparability between SIDS and NMSIDS cases and controls was limited, which may hamper the overall conclusion. At last, the included studies were small resulting in imprecision of estimates.

Conclusion

Despite many years of investigation, it is still impossible to conclude whether auditory system pathology is associated with SIDS. Four out of five histological studies indicate an association between the auditory system and SIDS; particularly abnormalities in the inferior colliculus. However, studies investigating the OAE or ABR and SIDS were inconsistent. The studies included in this review were generally of poor quality and future studies should focus on studying the association between auditory system pathology and SIDS in larger current study populations.

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Figure 1

Prisma flow chart showing the selection process of the systematic review

Contributorship statement

All authors had substantial contributions to this study. KD, MA, and TBH all contributed to the conception and design of the study. KD and MA screened titles and abstracts, subsequently performed full-text screening and data extraction with any disagreements resolved by discussion or by a third reviewer (TBH). KD drafted the systematic review with all authors (KD, MA, and TBH) revising it critically for important intellectual content prior to the submission. All authors approved the final

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version to be submitted. All authors had access to all data and take responsibility for the integrity of the reported findings.

Competing interests

None declared

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Data sharing statement

All data for this study is included in the article and the additional files.

Ethics statement

Ethical approval was not required as no animals or patients were involved. The systematic review included previously published human studies.

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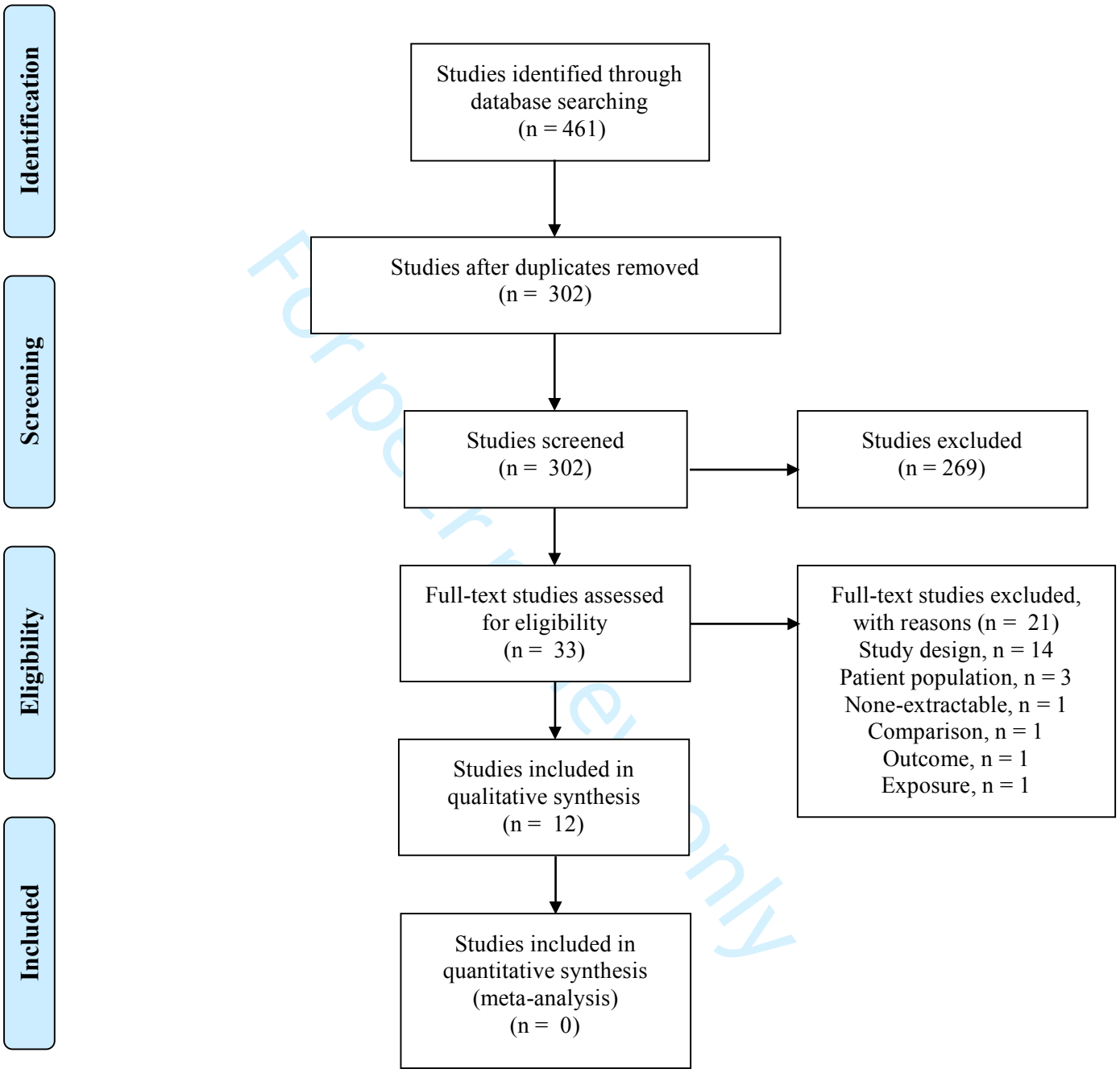
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Figure 1. Prisma flow chart showing the selection process of the systematic review



From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). *P*referred *R*eporting *I*tems for *S*ystematic Reviews and *M*eta-*A*nalyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit www.prisma-statement.org.

Additional file 1

The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

Katrine Dahl, Mads Andersen, Tine B. Henriksen

The full search strategy

Search strategy for Pubmed

1. "Sudden Infant Death" [Mesh]
2. sudden infant death
3. SIDS
4. cot death
5. crib death
6. apparent life threatening event
7. "Infantile Apparent Life-Threatening Event" [Mesh]
8. inner ear
9. auditory
10. "Auditory Pathways" [Mesh]
11. hearing
12. "Hearing" [Mesh]
13. "Hearing tests" [Mesh]
14. cochlear
15. "Cochlear Nucleus" [Mesh]
16. inferior colliculus
17. inferior colliculi
18. "Inferior Colliculi" [Mesh]
19. otoacoustic emission*
20. auditory brainstem response
21. "Evoked Potentials, Auditory, Brain Stem" [Mesh]
22. #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7
23. #8 OR #9 OR #10 OR #11 OR #12 OR #13 #14 #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21
24. #22 AND #23

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Search strategy for Embase

1. "sudden infant death syndrome"/exp
2. sudden infant death
3. SIDS
4. cot death
5. crib death
6. "near sudden infant death syndrome"/exp
7. apparent life threatening event
8. "apparent life threatening event"/exp
9. inner ear
10. "inner ear"/exp
11. auditory
12. "auditory system"/exp
13. hearing
14. "hearing test"/exp
15. cochlear
16. inferior colliculus
17. inferior colliculi
18. otoacoustic emission
19. otoacoustic emissions
20. auditory brainstem response
21. #1 OR #2 OR #3 OR #4 OR #5 OR #6 #7 OR #8
22. #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20
23. #21 AND #22

Search strategy for Web of Science

1. sudden infant death
2. SIDS
3. cot death
4. crib death
5. apparent life threatening event
6. inner ear

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7. auditory
8. hearing
9. cochlear
10. inferior colliculus
11. inferior colliculi
12. otoacoustic emission
13. otoacoustic emissions
14. auditory brainstem response
15. #1 OR #2 OR #3 OR #4 OR #5
16. #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14
17. #15 AND #16

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Additional file 2

The association between auditory system pathology and Sudden Infant Death Syndrome (SIDS): a systematic review

Katrine Dahl, Mads Andersen, Tine B. Henriksen

Excluded studies with reasons for exclusion

Title	Authors	Reason for exclusion
Is automated auditory brainstem response screening predictive of sudden infant death syndrome?	Brinsmead, T.L; Jardine, L; Thomas, D.	Study design
Neonatal otoacoustic emission screening and sudden infant death syndrome	Chan, R.S.Y.; McPherson, B; Zhang, V.W.	Study design
Inner ear and sudden infant death: Exploring implications for distubances in brainstem functions	Cohen, M.C.; Donn, T; Rivolta, M.; Cacciabue-Rivolta, D; Whitby, E; Rubens, D.	Study design
Brainstem auditory evoked-potentials in near-miss sudden infant death syndrome	Dorfman, L.J.; Gupta, P.R; Guilleminault, C.	Study design
Newborn hearing screen results for infants that died of SIDS in Michigan 2004-2006	Farquhar, L; Jennings, P.	Study design
Sudden infant death syndrome and possible relation to vestibular function	Farrimond, T.	Study design
Otoacoustic emissions does not currently have ability to detect SIDS	Hamill, T.; Lim, G.	Study design
Anatomo-pathologic study of the temporal bone in our 4 cases of sudden infant death	Josset, P.; Lecomte, D.; Omur, M.;	Study design

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	Chabolle, F.; Hannoun, L; Chouard, C.H.	
Auditory arousal thresholds of normal and near-miss SIDS infants	Kahn, A.; Picard, E.; Blum, D.	Outcome
Newborn hearing screens and SIDS	Krous, H.; Byard, R.W.	Study design
Histological study of the middle ear in the sudden infant death syndrome	Lecomte, D.; Josset, P.	Study design
Far-field auditory evoked potentials in near-miss sudden infant death syndrome	Luders, H.; Orlowski, J.P.; Dinner, D.S.	Study design
Abnormal brain stem potentials in infants with threatened sudden infant death syndrome	Nodar, R.H.; Lonsdale, D.; Orlowski, J.P.	Patient population
Abnormal brainstem auditory evoked potentials in infants with threatened sudden infant death syndrome	Orlowski, J.P.; Nodar, R.H.; Lonsdale, D.	Patient population
Prediction of apnea in siblings of sudden infant death syndrome victims using brain-stem auditory evoked-potentials	Orlowski, J.P.; Lonsdale, D.; Nodar, R.H.; Hunt; Strauss; Keenan; Rich.	None-extractable
Brainstem auditory evoked potentials in infants at risk of sudden infant death syndrome (SIDS) with and without aminophylline treatment	Reiterer, F.; Litscher, G.; Kurz, R.; Haidmayer, R.; Pfurtscheller, G.	Exposure
Medullary astrogliosis in sudden infant death syndrome varies with sleeping environment: Evidence for different	Spinelli, J.; Byard, R.W.; Van Den	Comparison

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mechanisms of death in alone versus co-sleepers?	Heuvel, C.; Collins-Praino, L.E.	
Brain-stem auditory evoked-potentials in sudden infant death syndrome (SIDS), near-miss-for-SIDS, and infant apnea syndromes	Stockard, J.J.; Hecox, K.	Study design
Hearing screening and sudden infant death syndrome	Stokowski, L.A.	Study design
Observations of otoconial membranes from human infants	Wright, C.G.; Hubbard, D.G.	Patient population
Absence of otoconia in a human infant	Wright, C.G.; Hubbard, D.G.; Graham, J.W.	Study design

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Reporting checklist for systematic review (with or without a meta-analysis).

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews

		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	1
Abstract			
Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	2
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	4
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	4
Methods			

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Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	5
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	6
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	Additional file 1
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and, if applicable, details of automation tools used in the process	6
Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	6
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	5
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	6
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used,	6

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		how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	
Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results ***The different studies were too heterogeneous to plan any calculation of effect measures and a narrative synthesis was planned.***	n/a
Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	6
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	7
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	7
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	7
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression) ***Not applicable.***	n/a
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results ***Not applicable.***	n/a
Reporting bias	#14	Describe any methods used to assess risk of bias due	7

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assessment		to missing results in a synthesis (arising from reporting biases)	
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	7
Results			
Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	8 Figure 1
Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	Additional file 2
Study characteristics	#17	Cite each included study and present its characteristics	8-11 incl. table 1
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	13-15 incl. table 2
Results of individual studies	#19	For all outcomes, present for each study (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	Table 1
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	13
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	Table 1
Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	Table 1

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Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results ***Not applicable.***	n/a
Risk of reporting biases in syntheses	#21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	16
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	16
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	16-17
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	18
Limitations of the review methods	#23c	Discuss any limitations of the review processes used	18
Implications	#23d	Discuss implications of the results for practice, policy, and future research	18
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	5
Registration and protocol	#24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	5
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol ***No amendments done.***	n/a
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	3

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Competing interests	#26	Declare any competing interests of review authors	3
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	19

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